UMM AL QURA UNIVERSITY

COLLEGE OF COMPUTING

FINAL YEAR PROJECT IN COMPUTER SCIENCE

Project: BVN Batal -Value Nest (بطل)

Project ID: CSAI-461-P1-F15

Team Name	Hero Builders
Team Logo	
Team Members	Hala Mohamed Jalal – [redacted] Khadijah Abdullah Gheliwi – [redacted] Shaimaa Mohamed Al-Kuzaee – [redacted] Mawaddah Mohammed Dehlawi – [redacted]
Project Leader	Mawaddah Mohammed Dehlawi – [redacted]
Project Supervisor	Dr. Elham Mahjoob Hassanain [redacted]
Start – end	22/8/2024 – 31/1/2025
Credit Hrs.	4 Hrs

Acknowledgment

First and foremost, we begin with expressing our profound gratitude to God Almighty for the endless blessings, guidance, and grace provided. God Almighty was our constant source of inspiration and strength, enabling us to persevere through challenges and work towards realizing our dreams. Our deepest appreciation goes to Dr. Elham Mahjoob Hassanain for her invaluable guidance, encouragement, and expert advice throughout the project. Her insightful feedback and unwavering support have significantly contributed to the success of this endeavor.

Moreover, we are sincerely thankful to our teammates, whose collaborative efforts, diverse perspectives, and insightful discussions enriched this project. It was through our collective dedication and teamwork that we were able to overcome obstacles and bring this vision to become a reality.

We are immensely grateful to our families for their steadfast encouragement, patience, and support throughout this journey. Their belief in us has been a pillar of strength during moments of difficulty and despair. This project would not have been possible without their unwavering love and motivation.

Finally, we dedicate this work to everyone who has inspired, supported, and motivated us along the way. Each contribution, big or small, has played a part in shaping this achievement, for which we are truly grateful.

Abstract

The fast digital transformation in this Era has made parent's duties in fostering positive habits and core values in their children a tremendous challenge. Children's immersions in digital technologies have made it a very lucrative method for helping parents to overcome these challenges through specialized apps. However, the lack of apps for Value enrooting particularly Arabic language apps makes it a rich field to invest in to build on relevant culturally based values.

Existing habit-tracking applications often target English-speaking audiences and, in most cases, lack culturally relevant content.

This project, BVN Batal – Value Nest, addresses this gap by creating an interactive value building through habit-tracking app designed for Arabic-speaking families. The primary objective is to encourage children to internalize essential values, such as honesty, cooperation, and responsibility, through gamification.

The **BVN** app provides a comprehensive system for integrating cultural values through daily tasks, reinforcing a reward system that allows parents to assign and monitor activities that encourage positive behaviors. The app features customizable rewards, progress tracking, and engaging virtual avatars, enhancing children's intrinsic motivation.

By actively involving parents in their child's habit and value formation journey, the app empowers them to effectively guide their child's development. It also leverages parental feedback to enhance user experience and ensure cultural relevance, bridging traditional values with modern technology.

Results have proven that the **BVN** app offers a sustainable framework for habit formation, fostering holistic child development while engaging them in meaningful activities. This innovative approach contributes to the digital parenting landscape by blending technology with cultural identity.

Keywords: interactive habit-tracking, value enrootment system, gamified task management, mobile app development, cross-platform development, parental control system, real-time progress tracking, UI/UX design for children, Arabic language interface app.

Table of Content

Acknowledgment *2*

Abstract *3*

Table of Figures *6*

List Of Tables *8*

Glossary *9*

Chapter I: Introduction *1*

1.0 Problem Definition **1**

1.1 Proposed Solution **1**

1.2 Aims and Objectives **2**

1.3 Data Collection **3**

Chapter II: System Analysis *5*

2.0 Data Requirements **5**

2.1 Services and Capabilities **5**

2.2 Requirements Specifications **6**

2.2.3 Non-Functional Requirements 10

Chapter III: System Design *12*

3.0 Context Diagram **12**

3.1 Data Flow Diagrams (DFD) **13**

3.1.1 level 1 Diagram 13

3.1.1.1 sign up (Registration) 14

3.1.1.2 sign in 14

3.1.1.3Add new child 15

3.1.1.4 Assign specific tasks 15

3.1.1.5 Rewards 16

3.1.1.6 Check tasks 17

3.1.1.7 Monitor Progress 18

3.1.1.8 Challenges 19

3.1.1.9 view inventory 20

3.3 Use Case Diagram **21**

3.4 Scenarios **22**

3.5 Class Diagram **28**

3.6 Activity Diagrams **29**

3.7 Sequence Diagrams **37**

Chapter IV: Prototype *46*
4.0 Home page: **47**
4.1 Registration page: **47**
4.2 Log-in page: **48**
4.3 Parent account page: **48**
4.4 Settings: **49**
4.5 Create a user for a child: **49**
4.6 Childs Overview page: **50**
4.7 Assign tasks: **50**
4.8 Give rewards: **53**
4.9 Initiate challenges: **54**
4.10 Child-Log in page: **55**
4.11 Childs home page: **56**
4.12 To do list page: **57**
4.13 Prayer tracker page: **58**
4.14 Challenges page: **59**
4.15 Progress page: **60**
4.16 Rewards inventory: **61**
Chapter V: Implementation Plan *63*
5.1 Development Environment and Tools **63**
5.2 System Architecture and Frameworks **63**
5.3 Database Implementation **63**
5.4 Core Functionalities **64**
5.5 Gamification Techniques **64**
5.6 User Interface (UI) and User Experience (UX) **64**
References *66*
Appendix *71*

Table of Figures

Figure 3.0: Context diagram for "Batal "System Architecture 12
Figure 3.1.1: DFD level 1 13
Figure 3.1.1.1: DFD child diagram - sign-up (Registration) 14
Figure 3.1.1.2: DFD child diagram - sign-in 14
Figure 3.1.1.3: DFD child diagram - Add new child 15
Figure 3.1.1.4: DFD child diagram - Assign specific task 15
Figure 3.1.1.5: DFD child diagram - Rewards 16
Figure 3.1.1.6: DFD child diagram - Check task 17
Figure 3.1.1.7: DFD child diagram - Monitor progress 18
Figure 3.1.1.8: DFD child diagram - challenges 19
Figure 3.1.1.9: DFD child diagram - View inventory 20
Figure 3.3: Use case diagram 21
Figure 3.5: Class diagram 28
Figure 3.6.1: Activity diagram - Sign-up / Sign-in 29
Figure 3.6.2: Activity diagram - Add new child 30
Figure 3.6.3: Activity diagram - Assign task 31
Figure 3.6.4: Activity diagram - Reward 32
Figure 3.6.5: Activity diagram - Check tasks 33
Figure 3.6.6: Activity diagram - Monitor progress 34
Figure 3.6.7: Activity diagram - Challenges 35
Figure 3.6.8: Activity diagram - view inventory 36
Figure 3.7.1: Sign up sequence diagram 37
Figure 3.7.2: Sign in sequence diagram 38
Figure 3.7.3: Add new child sequence diagram 39
Figure 3.7.4: Assign task sequence diagram 40
Figure 3.7.5: Rewards sequence diagram 41
Figure 3.7.6: Check task sequence diagram 42
Figure 3.7.7: Monitor progress sequence diagram 43
Figure 3.7.8: Challenges sequence diagram 44
Figure 3.7.9: View inventory sequence diagram 45
Figure 4.1: Interface – Home page 47
Figure 4.2: Interface – Registration page 47
Figure 4.3: Interface – Log-in page 48
Figure 4.5: Interface – Parent account page 48
Figure 4.5: Interface – Settings 49
Figure 4.6 Interface – Add child 49
Figure 4.6 Interface – Child's Overview Page 50
Figure 4.8: Interface – Assign tasks 50
Figure 4.8.1: Interface – Create task 51
Figure 4.8.2: Interface – Choose task 51
Figure 4.8.3: Interface – Choose task 52
Figure 4.9: Interface – Give rewards 53
Figure 4.10: Interface – Initiate challenge 54
Figure 4.12: Interface – Child home page 56

Figure 4.13: Interface - To-Do list page 57
Figure 4.14: Interface – Challenge page 59
Figure 4.15: Interface – Progress page 60

List Of Tables

Table 3.4.1: Use case Scenarios – sign-up 22
Table 3.4.2: Use case Scenarios – sign-in 22
Table 3.4.3: Use case Scenarios – Add new child 23
Table 3.4.4: Use case Scenarios – Assign task 23
Table 3.4.5: Use case Scenarios – Rewards 24
Table 3.4.6: Use case Scenarios – Check Task 24
Table 3.4.7: Use case Scenarios – Monitor progress 25
Table 3.4.8: Use case Scenarios – Challenges 26
Table 3.4.9: Use case Scenarios – View Inventory 26

Glossary

Acronyms and Technical Terms:

Term	Definition
BVN	**Batal Value Nest:** The project name, focusing on gamified habit-building for children aged 6-9 through positive tasks and values.
DFD	**Data Flow Diagram**
UML	**Unified Modeling Language**
SQL	**Structured Query Language**
UI	**User Interface**
FR	**Functional Requirements**
NFR	**Non-Functional Requirements:** System characteristics like usability, performance, and accessibility that enhance the user experience.
ID	**Identifier:** A unique value used to distinguish users (parents or children) and track their activities within the app.
Localization	The process of adapting software to meet the language, cultural, and regional requirements of users.

Chapter I: Introduction

In a time when digital technology infiltrates every part of daily life increasing the time spent on digital apps instead of spending this vital time bonding with their parents and learning the family value systems. one of the main reasons it has become more difficult for parents to build the essential values they would like to foster in their children's behaviors. The appeal of interactive digital applications frequently outweighs the effectiveness of conventional methods for forming habits in children. Even though the market is full of different digital applications for children to support families in the development of habits in different areas supporting their upbringing, there is a lack of Arabic language apps to support Arabic speaking families. The following sections provide a review of recent available digital habit-tracking apps which are examined forcing on how to support values-based development while tracking and reinforcing values through habit formation.

1.0 Problem Definition

In today's fast-paced digital world, core values in children and forming positive habits have become increasingly challenging for parents. Many traditional methods of habit formation lack engagement and fail to capture the interest of children who are accustomed to interactive digital experiences in all disciplines of their lives.

Existing habit-tracking apps like "Chore Monster" and " Habitica " have been successful in fostering routine-building among users but often fall short in offering personalized and value-driven experiences, especially in Arabic- language.

Moreover, there is a noticeable gap in the market for applications designed specifically for the age group between 6 and 9 that integrate culturally relevant values into the habit-forming process.

While platforms such as "Supers “have attempted to address this need, they often miss the connection between habit formation and value enrooting, particularly in an engaging, personalized manner. Given the increasing reliance on mobile applications as tools for children’s development, there exists a need for solutions which will not only foster positive habits but focuses on the integration of vital values into daily routines in an engaging and culturally relevant way.

1.1 Proposed Solution

The proposed system addresses the main issue of fostering the enrootment of a strong value system in children aged 6 to 9 by providing a culturally tailored habit-forming and tracking application. The app will be developed in Arabic with a user-friendly interface that aligns with the cultural values and norms of the region, ensuring that both the content and design are relatable and impactful. By integrating gamification elements such as rewards, badges, and interactive avatars, the app aims to motivate children to adopt and maintain positive habits tied to

core values like honesty, empathy, responsibility, cooperation, and respect. The system will allow parents to play an active role by assigning tasks, monitoring progress, and customizing rewards to suit their family's unique needs. The interface will follow a right-to-left (RTL) layout with culturally relevant visuals, color schemes, and fonts, ensuring usability and engagement for young users. The system's primary goal is to blend modern technology with traditional values, empowering families to guide their children's development in a meaningful and interactive way.

. Through the app, parents will be capable of assigning value-driven tasks which guide children to understand and practice these principles in their daily lives. By linking habit formation with value development, the app promotes intrinsic motivation, ensuring that children not only complete the tasks on hand but also internalize the values behind them. Children will earn points, badges, and other virtual rewards for completing tasks, to reinforce positive behavior and create a sense of achievement.

This approach encourages children to continue engaging with the app and their tasks. Parents play a crucial role in shaping their children's behavior, and this project, **BVN Batal - Value Nest** will attempt to support the parents by providing them with the capability o:

- Assign tasks and goals based on their child's progress and values.
- Monitor progress through detailed reports and updates.
- Customize rewards to fit the unique needs and values of their child.

Grasp the children's attention through constant achievement indicators on screen.

1.2 Aims and Objectives

The primary aim of this project is to assist parents in instilling noble human and Islamic values in their children, ensuring that these values are sustainable impactful even after the app is no longer in use. The main objectives are as follows:

1. **Value enrootment**: To provide a framework that helps parents embed essential Islamic and Nobel human values in their children's daily routines, building a strong moral foundation.
2. **Sustainable Habit Formation**: To design features that encourage children to build and maintain positive value related habits which persist over time, promoting long-term behavioral change.
3. **Engaging Learning Environment**: To create an interactive and enjoyable experience which captivates children's interest, making the habit-building process fun and educational.
4. **Intrinsic Motivation**: To cultivate a sense of internal motivation in children, encouraging them to adopt positive behaviors independently and fostering personal growth.
5. **Cultural Relevance**: To ensure that the app's content is culturally relevant and resonates with Arabic-speaking families, bridging the gap between traditional values and modern technology.

6. **Parental Guidance**: To equip parents with tools and resources that support their role in guiding their children to become noble and productive members of society, instilling values that contribute to shaping a responsible and compassionate generation.

By achieving these objectives, the aim is to empower children to develop strong habits and a high moral value system which will contribute to their personal development and the well-being of their communities.

1.3 Data Collection

To measure the need for the value habit tracker a parent survey was used as parents may have a more direct insight into their children's values and daily habits. Their feedback helped ensure the app meets real-world needs and address specific concerns. Additionally, involving parents allowed for the development of a tool which fulfils and support family-centered growth.

Surveys were chosen as the method of measurement as they offer a structured and efficient way to collect diverse feedback from many parents. This approach helped ensure that the app was built based on real life needs, also allowing for the identification of common patterns and concerns among families. A total of 100 surveys (Appendix 1.1) were conducted across a diverse group of parents, without targeting a specific region, ensuring broad representation. The full survey results may be reviewed in

Survey results (Appendix 1.1), offered valuable insights into parental preferences and challenges, shaping the design and goals of the BVN application.

The key findings may be summarized as follows:

1. **Usage of Digital Learning Applications:**
    - A significant proportion of parents, ranging from **33.0%** for daily use to **43.2%** for weekly use, integrate digital applications into their children's learning routines, with many using these tools regularly.
    - A smaller group of parents, **5.4%**, use such applications monthly, while **1.4%** use them rarely.
2. **Values Considered Important for Child Development:**
    - Core values highlighted by parents include respect, honesty, cooperation, responsibility, and empathy.
    - These align closely with the BVN app's goal of fostering positive habits and values in children.
3. **Desired Features in a Habit-Tracking Application:**
    - Parents prioritize features such as task assignment, rewards systems, progress tracking, interactive elements (e.g., avatars), and parental controls.

- The use of rewards is perceived as an effective approach to motivate children in habit formation and maintenance.

4. **Challenges in Instilling Values:**
    - Key challenges include effectively communicating values to children, managing external influences from peers and digital media, and maintaining children's focus and attention.
    - The BVN application aims to address these challenges by offering engaging content that promotes positive behavior and reinforces essential values.
5. **Strategies for Value Instilling:**
    - Parents employ a range of strategies, including storytelling, reward systems, modeling positive behavior, and providing consistent guidance and reinforcement.
    - The emphasis is placed on both theoretical discussions of values and practical demonstrations through parents' behavior.
6. **Importance of Arabic Language Availability:**
    - Most parents, **77%**, consider the availability of an Arabic-language application to be important or very important, underscoring the need for culturally relevant and accessible content.

The survey results highly backed up the need for a habit-tracking application that supports parents in effectively embedding values in their children. The BVN app aims to meet these needs through user-friendly features, reward mechanisms, and content tailored to the Arabic-speaking community.

BVN seeks to offer a tool that not only fosters enduring habits but also encourages internalized moral development through features like parental control, task customization, and rewards. To be precise, BVN Batal is positioned as a comprehensive platform that empowers parents to play a key part in their child's growth and development by addressing the ethical and practical needs of habit formation.

Chapter II: System Analysis

This chapter is organized to address both functional and non-functional requirements. Functional requirements outline how users will interact with the system and the results they can anticipate, as well as the fundamental features and services the program must provide. On the other side, non-functional requirements deal with important system characteristics that are essential to providing a flawless user experience, such as usability, performance, reliability, localization, maintainability, accessibility, and branding. This chapter serves as a foundation for understanding how the application will operate and the standards it must uphold to effectively support families in nurturing positive habits, all while providing an enjoyable experience that resonates with young users and their parents alike.

2.0 Data Requirements

The BVN application requires structured data to manage essential user information, including profiles, task assignments, rewards, and progress tracking. The system must ensure secure data handling, offering seamless user interactions for parents and children. This includes maintaining logs for task status reward distribution, and real-time progress updates. The app will also incorporate localization features to deliver an Arabic-friendly interface with culturally relevant content, fostering meaningful engagement and promoting value-based learning. Data management must support scalability to accommodate multiple users and maintain smooth performance over time.

To enhance accessibility and usability, the BVN application should be available on smartphones and tablets, ensuring quick access for both parents and children. It must run efficiently on iOS platforms, providing a responsive, user-friendly experience across various devices and screen sizes. This flexibility allows parents to assign tasks, review progress, and approve rewards on the go, while children can engage with the app throughout their day, completing tasks and earning rewards. The optimized system architecture ensures fast loading times, minimal resource consumption, and smooth performance, promoting regular use and maintaining user engagement across all devices.

2.1 Services and Capabilities

Through an interactive and entertaining platform, the application provides a full range of services aimed at helping parents and kids develop healthy habits. Children gain from entertaining and inspiring tools like star charts and customized pet avatars, while parents can oversee their children's accounts, assign tasks, and monitor progress thanks to the main features.

These services fall into two primary categories: firstly, the child side, which emphasizes participation and the formation of beneficial habits, and secondly, the parent side, which concentrates on administrative and motivational assistance.

Services offered to parents include creating an account, logging in and out, adding dependents (children), delegating tasks, tracking progress, and encouraging sibling rivalry through challenges. Parents can give their kids rewards for doing chores, using pre-made tasks or making their own, and tracking their progress.
With features such as daily activities, a star chart system to monitor progress, and a customizable pet avatar that changes according to the child's achievement, the application provides a very interesting experience for kids. Together, these components provide an enjoyable and engaging atmosphere that promotes the formation of positive habits in kids.

The key services include:

1. Account creation
2. Sign in / sign out
3. Add children (Dependents) - parent side
4. Assign tasks (use built-in tasks or create new ones) - parent side
5. Give rewards - parent side
6. Monitor progress - both sides
7. Challenges (between children under the same parent) - parent side
8. Engageable pet avatar - child side
9. Star chart - child side
10. Daily tasks - child side

2.2 Requirements Specifications

The system requirements for “BATAL” may be divided into functional and non-functional categories. Functional requirements outline the core features and capabilities the app must provide to meet user needs, such as tracking habits, rewarding achievements, and supporting Arabic language integration. Non-functional requirements define the app's performance, security, and usability standards to ensure a smooth and reliable user experience.

2.2.1.1 Account Creation

Parent Registration:
By entering basic details like their name, email address, and password, the parent can create an account. Parents can access and control the app's features after creating an account, which is secured by the system.

- The parent shall be able to create an account by providing the required information like name, email, and password.

2.2.1.2 Sign In / Sign Out

Sign In Process:
Children and parents use the same email address and password to log in. The system shows every user profile that is accessible after logging in, including the parent and any children who have registered.

- The parent and children shall sign in using the same email and password.
- After successful sign-in, the system shall display all user profiles (e.g., Parent, Child 1, Child 2, etc.) based on how many children are added.
    - The user shall select their profile to continue.
        - If the Parent profile is chosen:
            - The system shall display the parent dashboard, allowing access to features like adding dependents, assigning tasks, and monitoring progress.
        - If a Child profile is chosen:
            - The child shall be directed to their own profile, where they can access tasks, rewards, and progress tracking.

Sign Out:

- parent shall be able to sign out from their respective profiles.
- The system shall securely end the session upon sign-out.

2.2.1.3 Add Children (Dependents) - Parent Side

Add Dependent:
By entering basic details like the child's name and age, parents can add one or more dependent children to their profile. Additionally, they can choose a fixed avatar for each child from a list of predetermined options. By personalizing the app experience, this method makes it interesting and makes it simple for kids to recognize their profiles.

- The parent shall be able to add one or more children (dependents) to their profile.
- For each child, the parent shall provide information like name and age.
- The parent shall select a fixed avatar from a set of predefined options for the child.

Manage Child Profiles:

- The parent shall be able to edit the child’s details or change the avatar selection from the predefined set.
- The system shall maintain progress and habits for each child based on their profile and selected avatar.

2.2.1.4 Assign tasks (use built in tasks or create new ones) - parent side

The system shall allow parents to assign tasks to their children through the parent interface. Parents can choose from built-in task templates or create new custom tasks. The assigned tasks should be linked to the child's profile and be trackable within the app:

- The parent shall be able to view a list of built-in tasks (e.g., brushing teeth, completing homework, etc.) that can be assigned to the child.
- The parent shall be able to create custom tasks by specifying the task name, description, frequency (daily, weekly, etc.).
- The parent shall be able to assign tasks to a specific child by selecting their profile.
- The system shall send a notification to the child's app when a new task is assigned.
- The system shall allow the parent to set reminders or deadlines for each assigned task.
- The parent shall be able to view, edit, or delete both built-in and custom tasks after assigning them.

2.2.1.5 Give rewards - parent side

The system shall allow parents to assign rewards to their child's avatar based on the completion of tasks or positive behavior. The rewards should be customizable and visually reflected in the avatar's appearance or behavior within the app:

Built-in Rewards:

- The system shall automatically award coins to the child within the app for each completed task.
- The child can accumulate coins and use them within the app.

Custom Rewards by Parent:

- The parent shall be able to create custom rewards, assigning them to specific tasks or values that the child needs to complete.
- Upon the child's completion of the assigned task or value, the system will notify the child with a message detailing the custom reward.
- The parent shall be notified when the criteria for the custom reward are met and can confirm its release.

2.2.1.6 Monitor progress - both sides

The system shall allow both parents and children to monitor the progress of tasks in real-time. The parent can track the child's completion rate, milestones, and progress, while the child can view their task status and track their own progress toward rewards.

- The parent shall be able to view a detailed progress report showing the completion status of all assigned tasks for each child.

- The parent shall be able to filter tasks by date, category, or completion status (e.g., overdue, completed, in-progress).
- The parents shall receive notifications when tasks are completed or overdue.
- The child shall see which tasks are completed and which are still pending.

2.2.1.7 Challenges (between children under same parent) - parent side

A. **The user would be allowed to create challenges:** Parents could design and give competitive challenges to their children, like 'doing chores' or achieving certain goals.

B. **The user would be able to track the progress through:** A dashboard will provide the parent with the progress of each child through the challenge in a series of updates on task completion and points earned.

C. **Allow setting up rewards:** The ability can be provided to the parent to define a reward for the winner of each challenge-something like virtual items or badges.

D. **Performance comparison:** Relatively, through a leaderboard, the parent will be able to compare who has completed more activities or has scored higher among the children.

E. **Feedback from the user might be:** The parent can give feedback or encouragement to the children based on their performance in those challenges.

2.2.1.8 Engageable pet avatar - child side

a. **The user could earn rewards through the avatar:** By successfully completing tasks or challenges, the child can receive virtual rewards, such as food or accessories for the pet avatar, enhancing their engagement.

b. **The user could receive task reminders from the avatar:** The pet avatar will deliver engaging and motivational reminders to prompt the child to complete their assigned tasks in a fun and interactive way.

2.2.1.9 Star chart - child side

a. **The user could monitor progress using a star chart**: The child can track their progress through a star chart that gradually fills up as they complete tasks or challenges.

b. **The user could unlock new levels on the star chart:** By consistently completing tasks, the child can unlock additional levels or sections of the star chart, fostering long-term engagement and motivation.

c. **The user could display achievements:** The child can showcase earned stars or badges on the star chart, celebrating their accomplishments and milestones.

2.2.1.10 Daily tasks - child side

a. **The user could receive daily tasks:** The child will be presented with a list of daily tasks assigned by their parents, accessible through the avatar's interface.

b. **The user could receive feedback on task performance:** Upon successful task completion, the child will receive feedback or virtual praise from the avatar, reinforcing positive behavior.

c. **The user could track task streaks:** The child can monitor how many consecutive days they have completed their tasks, with opportunities to unlock new features or rewards for maintaining consistent streaks.

2.2.3 Non-Functional Requirements

The non-functional requirements focus on aspects such as performance, security, scalability, and usability, which are critical for the app's long-term success and user satisfaction. The following sections outline the specific non-functional criteria that the app must meet to maintain high-quality standards.

- **Usability**:

  1. The application must be user-friendly and intuitive, ensuring that both parents and children can navigate easily.
  2. The system should allow for minimal steps when performing basic functions, such as assigning tasks or viewing progress.

- **Performance**:

  The app should load within 3 seconds under normal network conditions and efficiently handle multiple concurrent users.

- **Reliability**:

  The application should have an uptime of 99.5% to ensure continuous availability for users.

- **Localization**:

1. The app must support Arabic language and cultural values, providing a contextually relevant experience for users.
2. All content and text must be tailored to **Arabic-speaking users**, ensuring a relatable experience.

- **Maintainability**:

  The system should be designed with the capacity for updates and maintenance, to allow feature enhancements when needed.

- **Accessibility**:

  1. The app should be accessible to users with disabilities, following best practices for inclusive design.
  2. The app interface should offer **voice-over support** and text size adjustability for special need users.

- **Branding**:

  The app should incorporate the "Hero Builders" team logo prominently, reinforcing brand identity.

The main features and prerequisites for the BVN Batal-Value Nest application were discussed to guarantee the app will help families instilling morals and good habits in children ages 6 to 9. While the non-functional requirements guarantee the app's usability, performance, and cultural relevance, the functional requirements specify crucial elements like task assignment, progress tracking, and rewards. When combined, these components offer a clear road map for developing a dependable and entertaining platform that encourages the development of values and long-term habits in an enjoyable and significant way.

Chapter III: System Design

This chapter discusses the design perspective, the application environment and the application architecture

3.0 Context Diagram

The context diagram effectively encapsulates the overall functionality and purpose of the application within its intended environment.

The Context diagram **Figure (3.0**) of the Batal-Value Nest system illustrates the system's interaction with its primary username, the parents and children in the diagram, the BVN system is depicted as a central entity that interfaces directly with both user groups. Parents can engage with the system to create accounts, assign tasks, and monitor their children's progress, while children interact with the system to complete tasks and earn rewards.

This visual representation emphasizes the roles of each user and the flow of information between them and the system, highlighting how the Batal application facilitates positive habit formation and value instillation in a user-friendly manner.

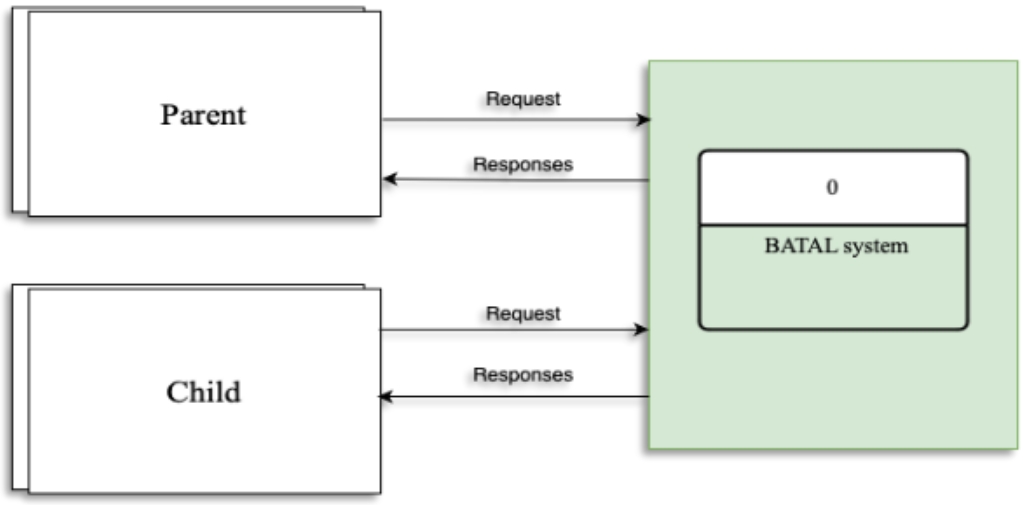


Figure 3.0: "Batal "System Architecture Context diagram

3.1 Data Flow Diagrams (DFD)

The following sections will break down the flow of data between the primary elements, such as user inputs, habit tracking, reward generation, and progress monitoring, offering insight into the BATAL system's structure and behavior.

3.1.1 level 1 Diagram

The Level 1 Data Flow Diagram (DFD) in **Figure (3.1.1)** provides a comprehensive overview of the system's processes and how data flows between the user (parent) and various components. It includes key processes such as "Sign Up", where the parent submits their information through a registration form, and "Verify Entered Data," which checks the accuracy of the submitted details. Upon successful verification, the system initiates the "Send Verification Email" process, sending a confirmation email to the parent's registered address. The diagram highlights the "Parent Table," a data store where all registered information is kept. Arrows indicate the flow of data between these processes and the database, emphasizing the seamless interaction between the user and the system. Overall, the Level 1 DFD effectively illustrates the core functions related to user registration and verification, laying the groundwork for understanding the system's architecture and user engagement.

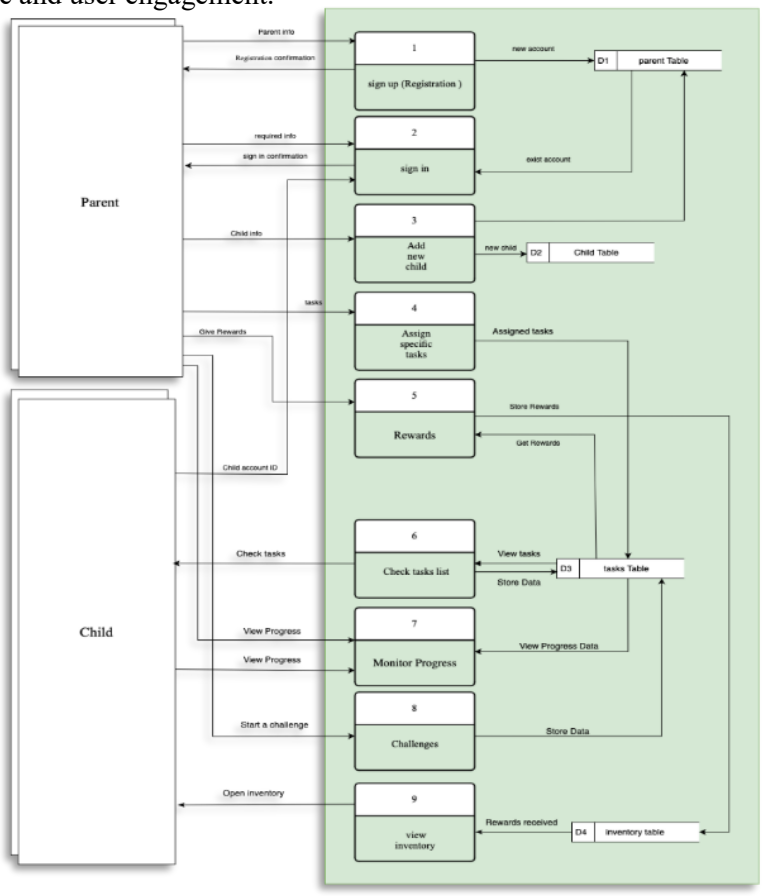


Figure 3.1.1: BVN - DFD level 1

3.1.1.1 sign up (Registration)

The child diagram for sign-up in the system outlines the process through which the parent entity can register parent information, as shown in **Figure (3.1.1.1).** The process (1.1) "show sign-up form" receives input from the parent, including parent information such as name and details. This information is passed to process (1.2) "Verify the entered data," where the system verifies the entered data. If the data is valid, it will be saved in the "parent table." If an error is detected, an error message is returned to the parent entity for correction. Upon successful verification and account creation, process (1.3) "send verification email" is triggered, where a verification email is sent to the parent's registered email address.

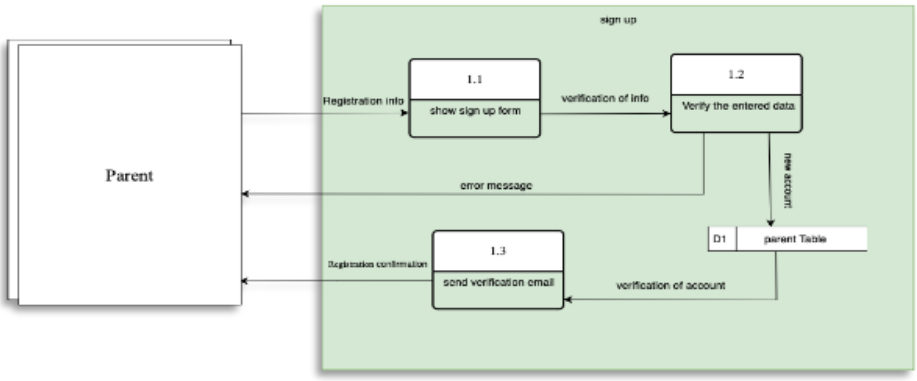


Figure 3.1.1.1: BVN-DFD child diagram - sign-up (Registration)

3.1.1.2 sign in

The child diagram for sign-in in the system in **Figure (3.1.1.2)** outlines the process through which the parent entity can log into the system, as shown in figure (4). The process (2.1) "show sign-in form" receives input from the parent, including required information such as the parent's email and password. This information is passed to process (2.2) "Verify the entered data," where the system checks the credentials against the existing account in the "parent table." If the credentials are valid, the parent's account is confirmed, and process (2.3) "show Users page" is triggered, the system displays a list of users (typically the children) associated with the parent's account. And the parents or children can select their account. (2.4) "Show profile page" is triggered, allowing the parents and children to access their profile page.

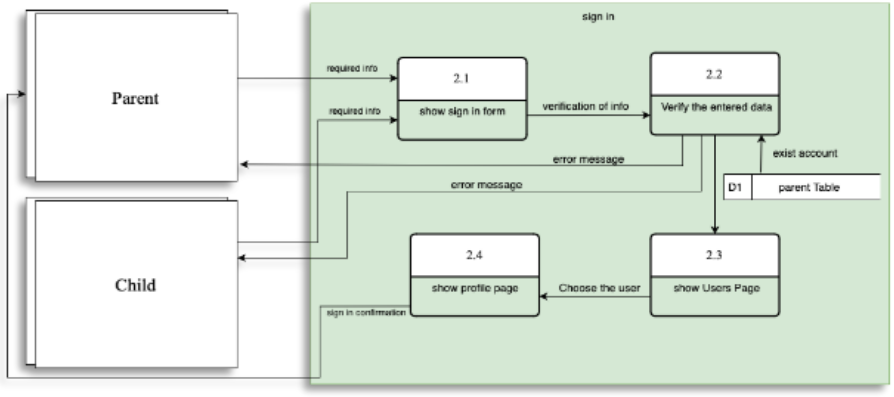


Figure 3.1.1.2: BVN- DFD child diagram - sign-in

3.1.1.3Add new child

The child diagram for adding a new child **Figure (3.1.1.3)** outlines the process through which the parent entity can register child information, as shown in the figure. The process (3.1) "Enter Child Details" receives input from the parent, including the child's information, such as name and details. This information is passed to process (3.2) "Validate Child Info," where the system checks the validity of the entered details. If an error is detected, an error message is returned to the parent entity. Otherwise, the process moves to (3.3) "Save Child Data" to store the validated information into the (D1: Parent Table) and ((D2: Child Table)) databases. Upon successful completion, the updated data is stored, and the process concludes.

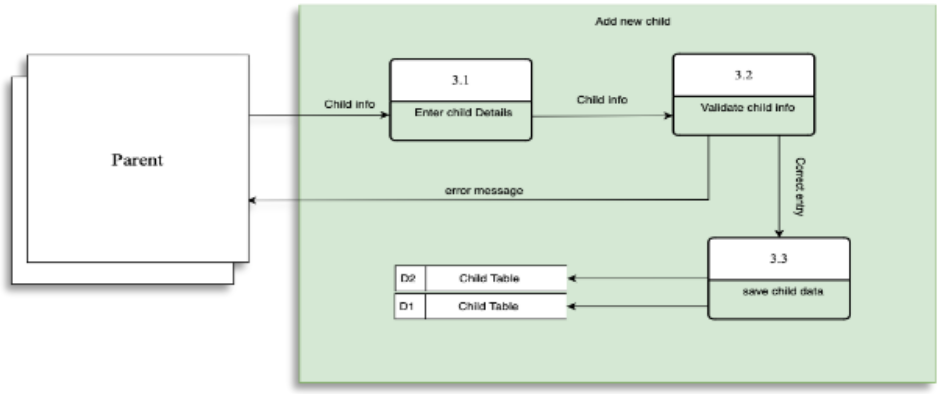


Figure 3.1.1.3: BVN- DFD child diagram - Add new child

3.1.1.4 Assign specific tasks

The child diagram in **Figure (3.1.1.4)** is for assigning specific tasks and illustrates how the parent entity can assign tasks to a child, as shown in the figure. The process begins with the parent directly interacting with (4.2) "Assign Tasks," where the parent selects the child to assign the tasks. Before this, the parent may create tasks through process (4.1) "Create Custom Tasks." The tasks are then stored in the (02: Child Table) and (03: Tasks Table) for reference. After assigning tasks, the parent proceeds to process (4.3) "Set Deadlines," where they can specify deadlines for each task. These deadlines are also saved in the (03: Tasks Table). This allows the parent to effectively manage, and track tasks assigned to the child.

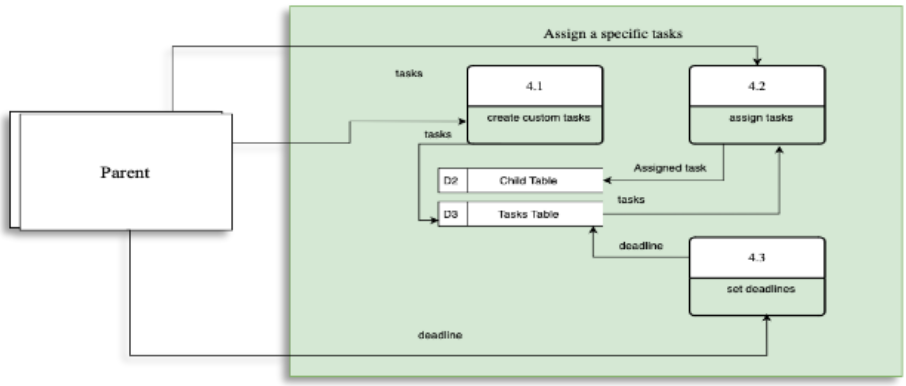


Figure 3.1.1.4: BVN -DFD child diagram - Assign specific task

3.1.1.5 Rewards

Figure (3.1.1.5) outlines the process by which parents create and assign rewards to their children within the app. A parent can create a custom reward, which is then linked to specific tasks. When a child checks their tasks and completes one, they can either win a custom reward or a built-in reward provided by the system. The system then stores this achievement in the inventory table. The child can view their earned rewards, offering motivation and reinforcing positive behavior. The structured flow between creating, assigning, and awarding both custom and built-in rewards ensures that the app's features are both interactive and goal oriented.

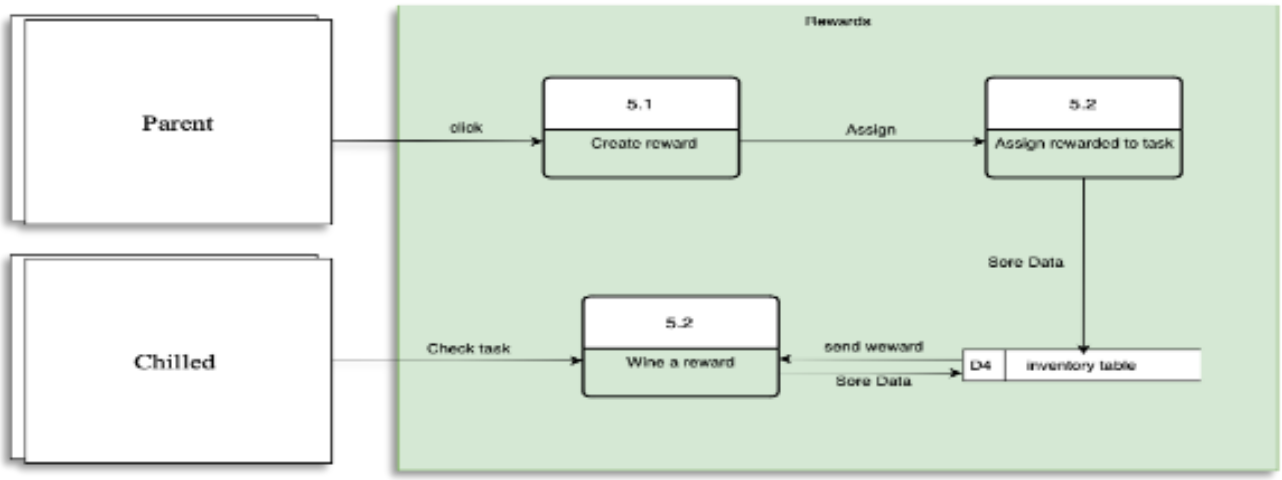


Figure 3.1.1.5: BVN- DFD child diagram - Rewards

3.1.1.6 Check tasks

The "Check Task" diagram in **Figure (3.1.1.6)** outlines the process through which a child can review and manage their assigned tasks within the system. It begins with the child accessing the task management interface, where a list of tasks is displayed. The child can select a specific task to view its details, including deadlines and status. If the task is overdue or requires updates, the system prompts the child with notifications. Additionally, the child has the option to mark tasks as complete, ensuring that task management is efficient and up to date. This process enhances user engagement and accountability by providing clear visibility into task progress and requirements.

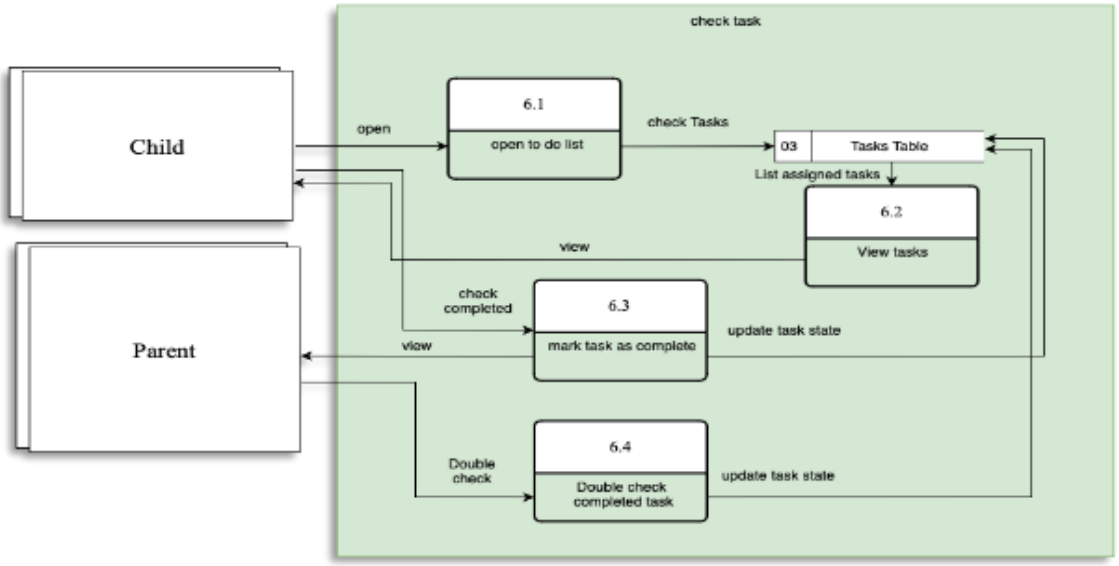


Figure 3.1.1.6: BVN- DFD child diagram - Check task

3.1.1.7 Monitor Progress

This diagram in **Figure (3.1.1.7)** details how both the parent and child can monitor the child's progress within the app. The parent can request data about the child's task completion, which is then retrieved from the tasks table. The system will compute the child's progress and visualizes it in charts. Both parents and children can view these visualizations, allowing them to understand how well the child is doing and where improvements can be made. This feature enhances transparency and keeps both parties informed about habit development.

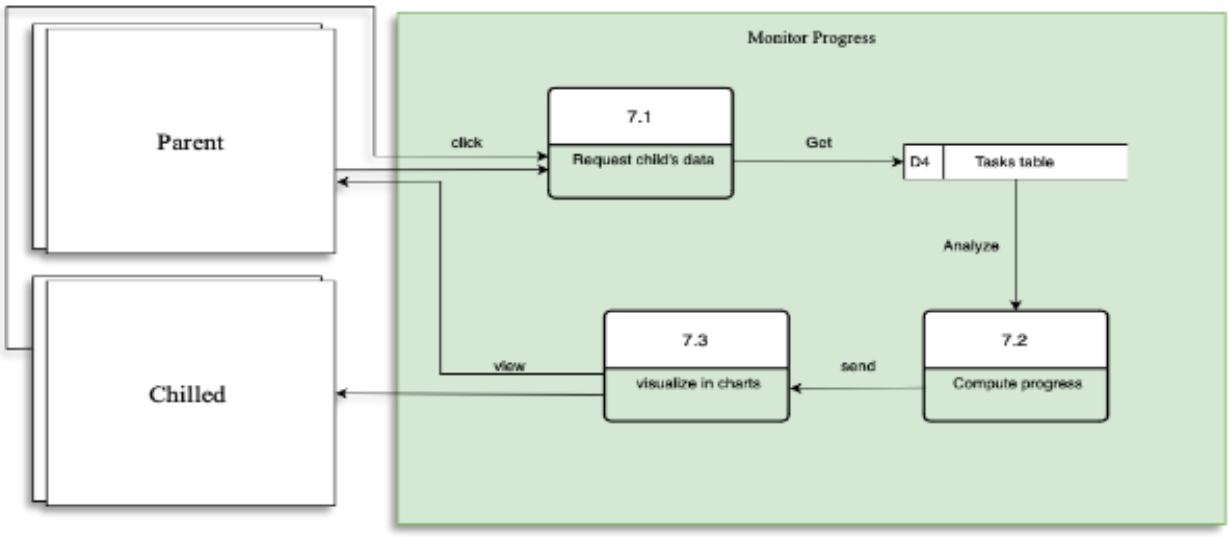


Figure 3.1.1.7: BVN- DFD child diagram - Monitor progress

3.1.1.8 Challenges

The challenges diagram in F**igure (3.1.1.8)** represents the process for creating and managing challenges in the app. The parent can initiate challenges by selecting specific children, and these challenges are stored in the tasks table. When a child is notified of a new challenge, they can view it as a message in their app. This encourages children to engage in friendly competition, motivating them to achieve specific goals and improve their habits as tracked by the app.

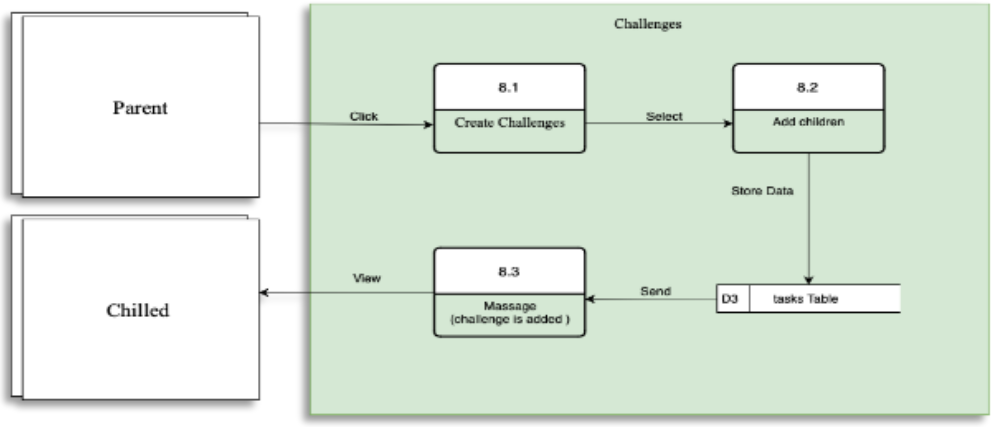


Figure 3.1.1.8: BVN- DFD child diagram - challenges

3.1.1.9 view inventory

The "View Inventory" child diagram in **Figure (3.1.1.9)** details the process through which a child can access and review all received rewards within the system. It begins with the child logging into their account and navigating to the rewards inventory section. Here, the child can see a comprehensive list of rewards they have earned, including details such as the reward name, description, and the date received. By selecting a specific reward, the child can view additional information, such as how it can be redeemed or used. This process enhances the child's engagement and motivation by providing clear visibility into their achievements and available rewards, fostering a sense of accomplishment and excitement.

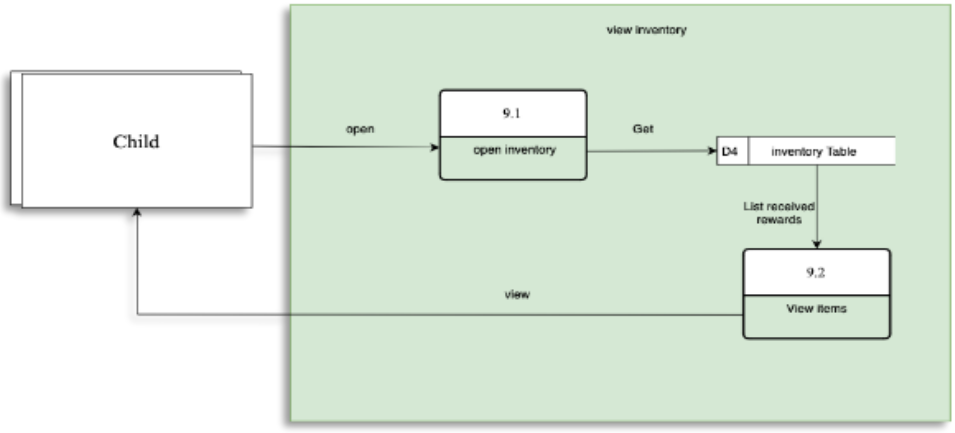


Figure 3.1.1.9: BVN -DFD child diagram - View inventory

3.3 Use Case Diagram

The use case diagram Figure (3.3) outlines the primary interactions between parents and children within the app. Parents can register, sign in, add children, assign tasks, and manage rewards. Children, on the other hand, can sign in, check task lists, engage with their pet avatars, and view inventory. The diagram also includes "Verify Password" and "Forgot Password" functionalities for secure access. Additionally, it shows the parent approval step when a child signs in using a generated code, ensuring secure access and parental control over the child's activities within the app.

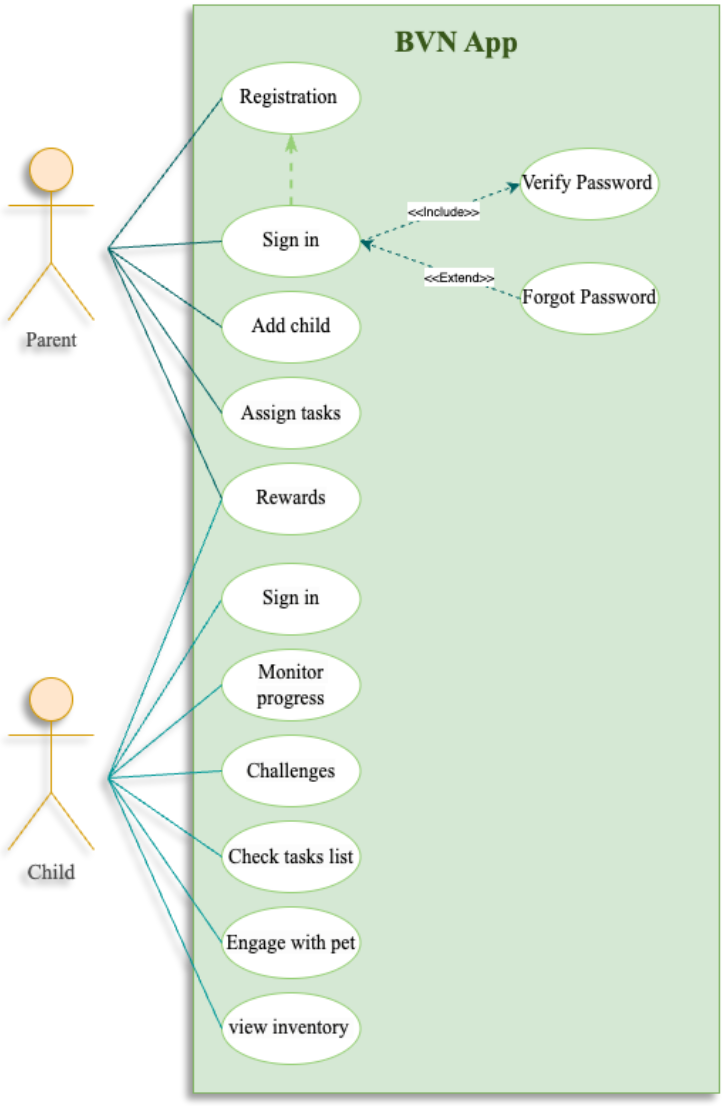


Figure 3.3: BVN- Use case diagram

3.4 Scenarios

Table 3.4.1: Use case Scenarios – sign-up.

sign-up	
Actors	Parent
Description	which a parent registers a new account in the system to access the application's features. The process includes filling out a sign-up form, verifying the entered information, sending a verification email, and confirming the registration.
Data	1. Parent's name (first Name – last Name) 2. Parent's ID 3. Parent's email address 4. Parent's phone number 5. Classification (father or mother) 6. Password
Stimulus	1. The parent initiates the sign-up process by navigating to the registration page and providing their details in the sign-up form. 2. The system responds by verifying the entered data (email format, password strength, etc.).
Respond	1. If the entered data is valid, the system saves the information to the "parent table" and triggers the sending of a verification email to the provided email address. 2. If there is an error in the entered data, an error message is displayed, prompting the parent to correct the information. 3. Once the email is verified through the confirmation link, the parent account is marked as fully verified, and the parent is able to access the system.
Comments	The verification email ensures that the parent's email address is valid, preventing the creation of fake or duplicate accounts.

Table 3.4.2: Use case Scenarios – sign-in

sign-in	
Actors	Parent, child
Description	The parent must provide their registered credentials (email and password) to gain access to the app's features. The system verifies the entered data, checks if the account exists, and grants access if the credentials are correct.
Data	1. Parent's email address 2. password
Stimulus	1. The parent initiates the sign-in process by navigating to the sign-in page and entering their credentials (email and password). 2. The system responds by verifying the provided data and checking the existence of the account in the parent database.
Respond	1. If the credentials are valid, the system grants access to the parent and redirects them to their profile page.

	2. If the credentials are incorrect or the account does not exist, an error message is displayed, asking the parent to re-enter valid credentials or sign up if they do not have an account.
Comments	The system should provide clear feedback to the parent when sign-in fails, with messages like "Invalid email or password" without specifying which field is incorrect to protect security.

Table 3.4.3: Use case Scenarios – Add new child

Add new child	
Actors	Parent
Description	The parent wants to add a new child's details into the system. This process involves inputting the child's information, validating it, and saving it in the system.
Data	Child's name and age.
Stimulus	The parent initiates the "Add New Child" process by selecting the option from the system.
Respond	1. The system prompts the parent to enter the child's details. 2. The system validates the entered information. 3. If correct, the child's details are saved into the child database. If incorrect, an error message is displayed. 4. The parent is notified that the child has been added successfully.
Comments	This process ensures that accurate data is entered into the system. The validation step is critical to avoid incorrect information.

Table 3.4.4: Use case Scenarios – Assign task

Assign task	
Actors	Parent
Description	The parent assigns tasks to a child, specifying deadlines and task details. This process includes creating custom tasks, assigning them to the child, and setting deadlines.
Data	Task description, task details, task deadline, and child's ID.
Stimulus	The parent selects the "Assign Task" option for a specific child.
Respond	1. The system prompts the parent to create a new custom task. 2. The task is assigned to the selected child. 3. The parent sets a deadline for the task. 4. The system saves the assigned task in the task table and links it to the child's profile. 5. The parent is notified that the task has been successfully assigned.
Comments	This process helps manage the child's tasks efficiently by allowing the parent to create custom tasks and set appropriate deadlines.

Table 3.4.5: Use case Scenarios – Rewards

Rewards	
Actors	Parent, Child, System
Description	The parent creates custom rewards or selects built-in rewards for tasks assigned to their child. However, even if the parent does not assign any reward to a task, the system automatically gives a built-in reward (such as coins or badges) when the child completes the task. These rewards are stored in the child's inventory.
Data	1. Task completion status 2. Reward details (custom or built-in) 3. Child's inventory
Stimulus	1. The child completes a task assigned by the parent or system. 1. If the parent has assigned a reward, the system links the reward to the task. 2. If no reward is assigned by the parent, the system automatically assigns a built-in reward.
Respond	1. The system verifies the task's completion. 2. If a custom reward is assigned by the parent, it is given to the child. 3. If no custom reward is assigned, the system automatically gives the child a built-in reward. 4. The child can view their earned rewards in their inventory, and the parent is notified of the successful task completion.
Comments	The automatic reward system ensures that the child remains motivated to complete tasks, even without explicit parental intervention. This process fosters continued engagement and positive behavior by rewarding children for their efforts in a consistent manner.

Table 3.4.6: Use case Scenarios – Check Task

Check task	
Actors	Child, Parent
Description	The child opens their to-do list to view the assigned tasks. After completing a task, the child marks it as completed. The parent can double-check the completed task to confirm if it was done correctly. The system updates the task state accordingly.
Data	1. Task data 2. Child's ID 3. Parent's ID 4. Task state (completed/not completed)

Stimulus	1. The child opens the to-do list to check assigned tasks. 2. The child marks a task as completed after finishing it 3. The parent opens the child's to do list 4. The parent double checks the completed tasks
Respond	1. The system retrieves the assigned tasks from the **Tasks Table** and displays them in the to-do list. 2. When the child marks a task as completed, the system updates the task's state in the **Tasks Table**. 3. The parent can double-check the completed task and either confirm it or send feedback, after which the system updates the task state again if necessary.
Comments	This process ensures that the child is honest and responsible for marking tasks as completed, while the parent maintains oversight by double-checking and validating task completion.

Table 3.4.7: Use case Scenarios – Monitor progress

Monitor progress	
Actors	Parent, Child
Description	The parent requests to view the child's progress on assigned tasks. The system computes and visualizes the progress in charts.
Data	1. Parent's ID 2. Child's ID 3. Child's task data
Stimulus	1. The parent clicks the "child's progress" option. 2. The system retrieves the child's task data and analyzes it to compute progress.
Respond	1. The system computes the child's progress based on task completion and sends the results to the visualization module. 2. The parent and child can both view the visualized progress in charts.
Comments	This process allows parents to monitor their child's progress easily and gain insights through visual charts.

Table 3.4.8: Use case Scenarios – Challenges

Challenges	
Actors	Parent, Child
Description	The parent creates challenges for the children, which can be competitive or cooperative, depending on the nature of the challenge. The children can view the challenge details and track their progress, encouraging engagement and habit formation.
Data	1. Challenge details (goal, deadline, rewards) 2. Child's progress data 3. Challenge leaderboard (if competitive)
Stimulus	1. The parent initiates a challenge creation by defining the goal and assigning the challenge to one or more children. 2. The system notifies the children that a new challenge has been added.
Respond	1. The children engage in the challenge, tracking their progress as they complete the assigned tasks. 2. The system updates the challenge progress in real-time, allowing both parents and children to view progress. 3. Upon completing the challenge, the child receives the assigned reward, and their progress is recorded.
Comments	Challenges create a sense of competition or teamwork between children, fostering engagement in tasks and helping them achieve specific goals. Parents can view the leaderboard to track each child's performance.

Table 3.4.9: Use case Scenarios – View Inventory

View inventory	
Actors	Child
Description	The child opens the inventory to view the rewards they have received from completing tasks.
Data	1. Child's ID 2. Inventory data (rewards/items)
Stimulus	1. The child initiates the process by selecting "open inventory." 2. The system retrieves the inventory data associated with the child's ID
Respond	1. If the child has valid rewards or items, the system lists the received rewards in the inventory.

	2. If the child has no items, a message is displayed indicating that the inventory is empty.
Comments	The inventory table ensures that all received rewards are recorded and can be viewed by the child upon request.

3.5 Class Diagram

The class diagram in **figure (3.5)** presented in the image demonstrates the interaction between the key entities in a task management and reward system for children. The "Parent" class is central, with attributes like FirstName, LastName, Email, Password, and ParentID, and methods for signing up, signing in, adding children, assigning tasks, giving rewards, and monitoring progress. The "Child" class, associated with the parent, includes attributes like ChildName, ChildID, ParentID, and level, alongside methods for marking tasks, viewing inventory, and challenges. The child is linked to "Task" and "Challenges" classes, which manage task details, repetition, deadlines, and state. The system also tracks rewards through an "Inventory" class, connecting children to various rewards, which are defined in the "Reward" class, offering both built-in and custom options. Overall, the diagram illustrates a robust relationship between tasks, rewards, and parental oversight for managing children's progress.

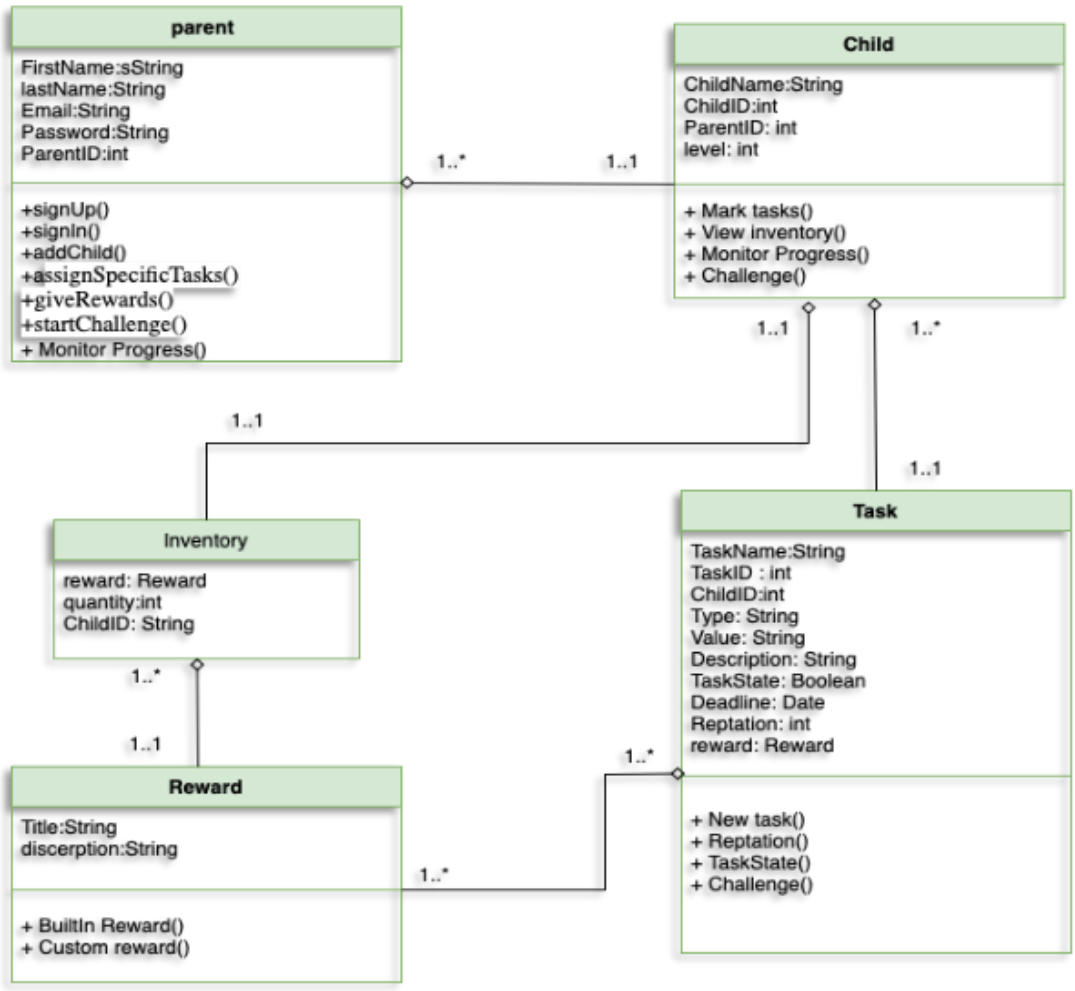


Figure 3.5: BVN -Class diagram

3.6 Activity Diagrams

3.6.1 sign up (Registration) / sign in

Figure (3.14) shows the flow for both the sign-up and sign-in processes. The sign-up process begins with the parent entering their information. The system then verifies the entered data. If the data is valid, the process proceeds to display the profile page. If there are errors, the parent is prompted to re-enter their information. The sign-in process starts similarly with the parents and child entering their credentials. The system verifies if the account exists. If the account exists, the user is directed to the user page, followed by the profile page. If there is an error in verifying the account, the process loops back to prompt the parent to correct the information. Both flows end with the display of the profile page.

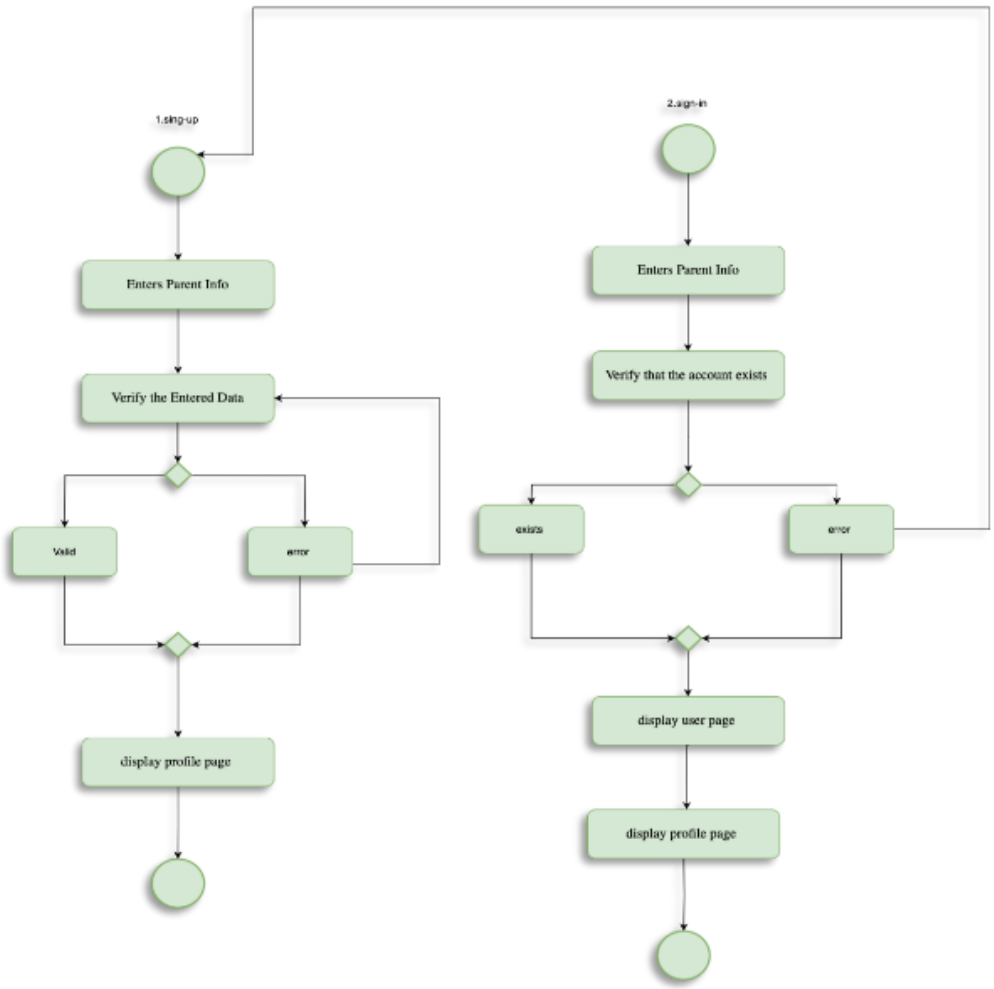


Figure 3.6.1: BVN -Activity diagram - Sign-up / Sign-in

3.6.2 Add new child

The add new child activity diagram in **figure (3.6.2)** outlines the steps involved in registering a new child. The process starts with the "Enter Child Details" activity, where the parent provides the child's information. Following this, a decision point checks the validity of the entered details. If the information is correct, the child's details are saved into the database. If the entry is invalid, an error is displayed, prompting the parent to correct the information before proceeding. This structure ensures that only valid child details are saved, while guiding the parent through necessary corrections if errors occur.

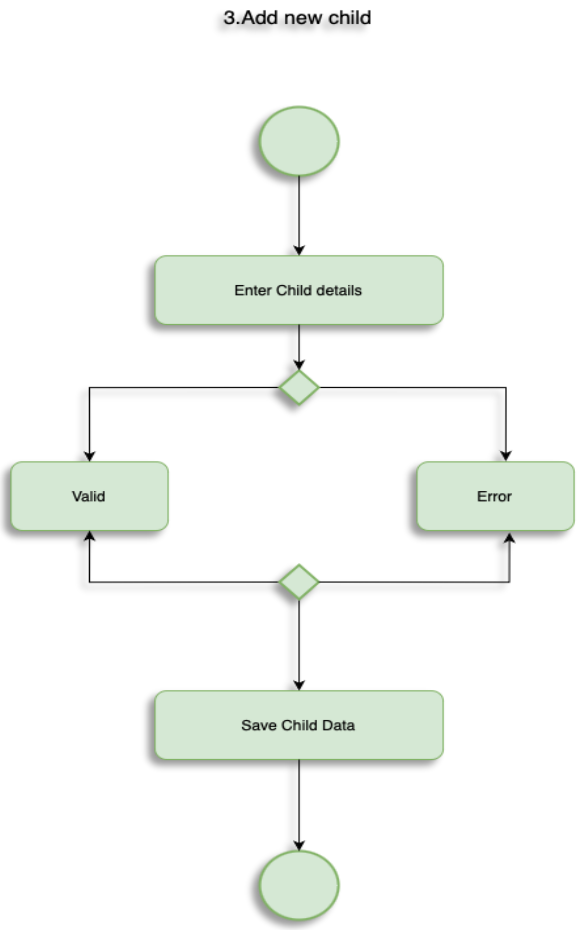


Figure 3.6.2: BVN- Activity diagram - Add new child

3.6.3 Assign specific tasks The assign task activity diagram in **figure (3.6.3)** outlines the process by which a parent can assign tasks to their child. The activity begins with a decision point where the parent chooses to either assign an already existing task or create a custom one. After making this choice, the flow moves to the "Set Deadline" activity. Next, the parent faces another decision: whether to add a reward for completing the task. Once these choices are made, the task details are saved to the database. This structure allows parents to either select from predefined tasks or create new ones, customize deadlines, and optionally offer rewards to motivate their children.

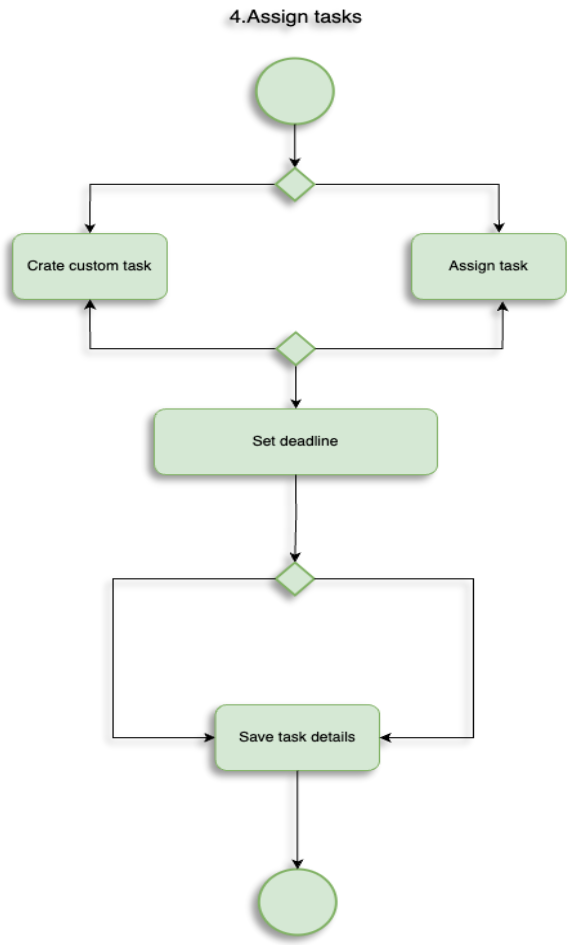


Figure 3.6.3: BVN -Activity diagram - Assign task

3.6.4 Create Rewards

This activity diagram in **Figure (3.6.4)** illustrates the process of assigning rewards to tasks. The parent begins by selecting a task for the child to complete. Afterward, the parent can either create a new custom reward or choose from a list of built-in rewards provided by the system. Once the reward is selected, it is saved and linked to the specific task. Upon task completion, the reward will be stored in the child's inventory. This flow ensures a flexible reward assignment system, allowing parents to customize rewards while also providing built-in options for efficiency.

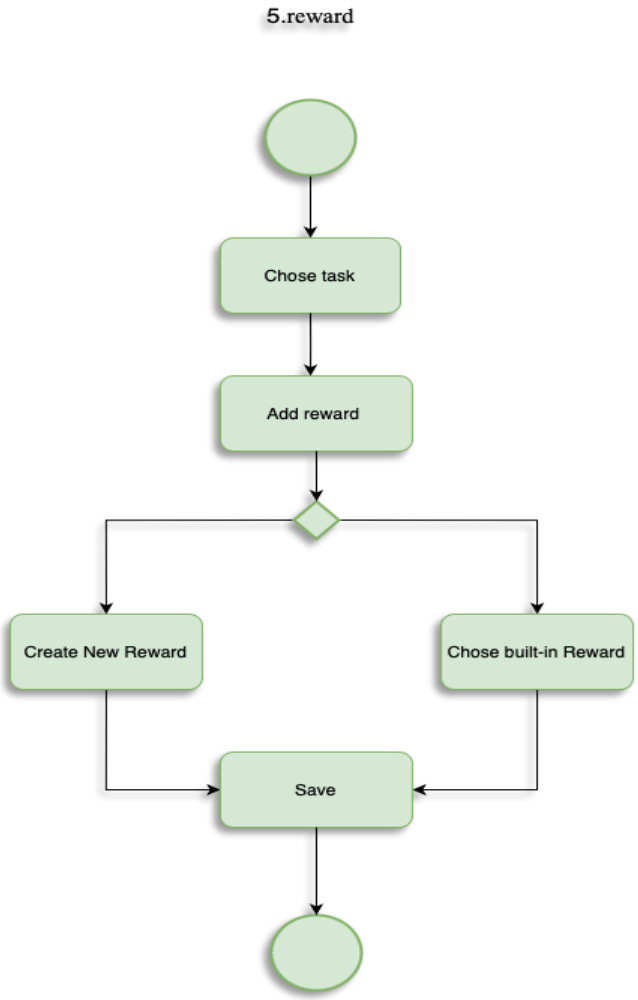


Figure 3.6.4: BVN- Activity diagram - Reward

3.6.5 Check tasks

This diagram in **figure (3.6.5)** outlines the process when a child or parent opens the to do list. The user views tasks and marks them as completed, triggering a status update. The parent double checks the completed task before the system updates it again and grants a reward to the child. This ensures both automatic and parent-verified completion.

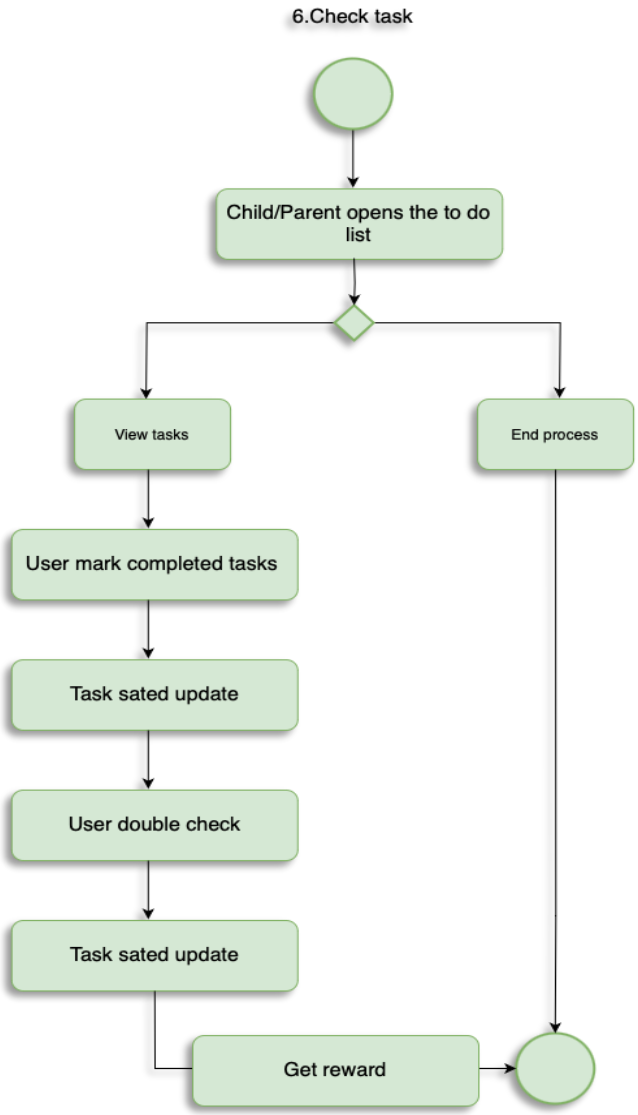


Figure 3.6.5: BVN -Activity diagram - Check tasks

3.6.6 Monitor Progress

The **Figure (3.6.6)** shows the flow of how a child or parent views the child's progress. After opening the progress interface, the system retrieves the child's data, computes the progress, and presents the results using charts. The visualized data is then sent back to the user for review.

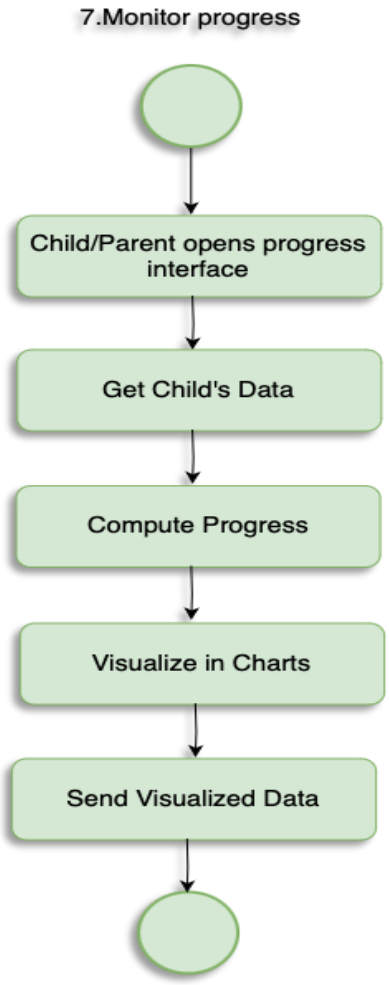


Figure 3.6.6: BVN - Activity diagram - Monitor progress

3.6.7 Challenges

The challenges activity diagram in figure (3.6.7) outlines the process of creating and managing challenges within the app. The parent begins by creating a new challenge and deciding whether it will be competitive or cooperative. After selecting the type of challenge, the parent sets a deadline for its completion. The option to add a reward for completing the challenge is available, and once everything is configured, the challenge is saved. This structure allows parents to encourage their children to engage in goal-oriented activities through competition or cooperation, fostering both teamwork and motivation.

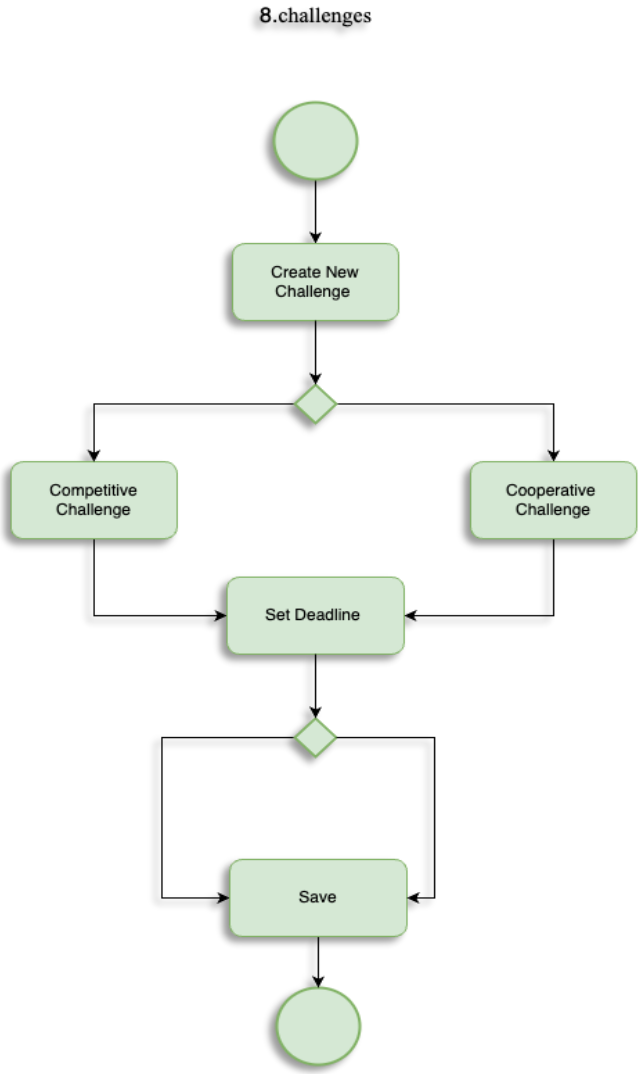


Figure 3.6.7: BVN -Activity diagram - Challenges

3.6.8 view inventory

In **Figure (2.6.8)**, the child opens the inventory interface to view the rewards they have earned. The system fetches inventory data, lists the received rewards, and allows the child to view items in their collection.

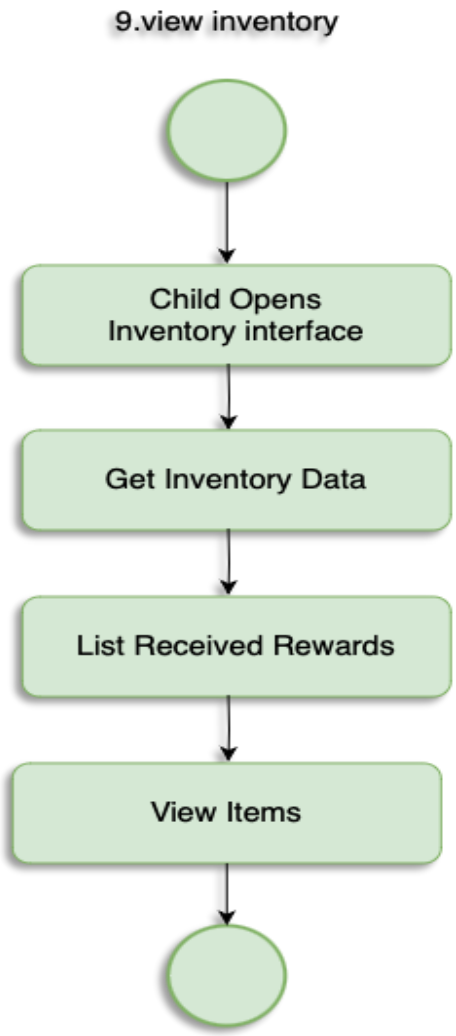


Figure 3.6.8: BVN- Activity diagram - view inventory

3.7 Sequence Diagrams

3.7.1 Sign up (Registration)

The sequence diagram for sign up **figure (3.7.1)** outlines the steps through which the **parent** inputs their name, email, and password into the system. These inputs are sent to the **BVN system**, which then verifies the email and password with the **verification system**. If verification succeeds, an email is sent back to the BVN system. If the input is valid, a new account is added to the database via the addNewAccount () function. If the input is invalid, an error message is displayed to the parent.

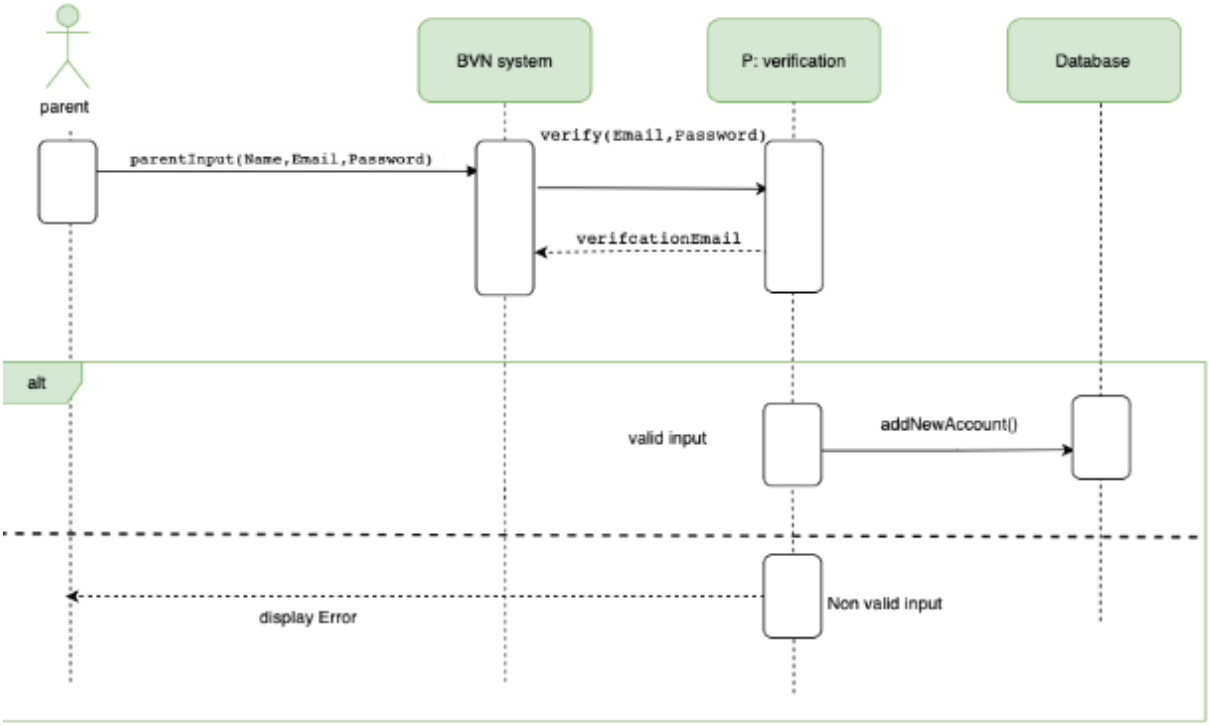


Figure 3.7.1: BVN- Sign up sequence diagram

3.7.2 Sign in

The sequence diagram for sign in **Figure (3.7.2)** outlines the steps through which the **user** submits their email and password to the system. The system forwards this information to the **BVN system**, which verifies the credentials with the **verification system**. If the account exists, the system retrieves it and shows the user's page. Valid input results in the system querying the database for the account, while invalid input triggers an error display.

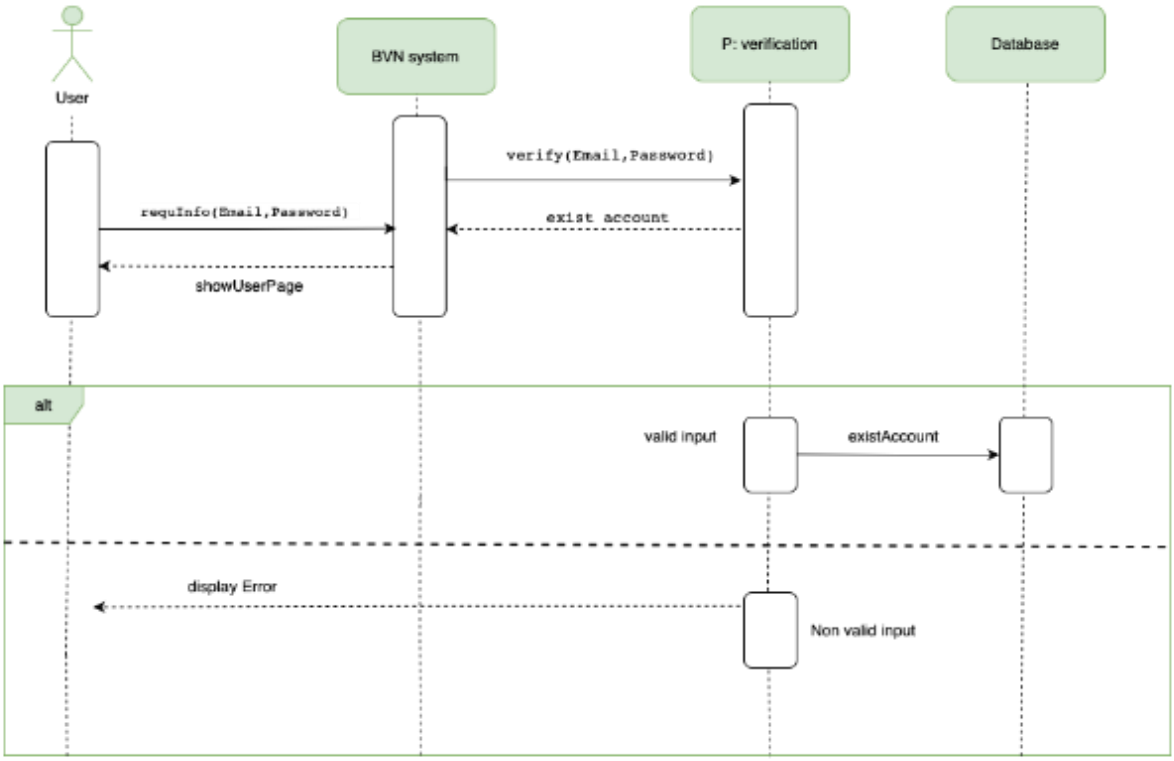


Figure 3.7.2: BVN -Sign in sequence diagram

3.7.3 Add new child

The sequence diagram for adding a new child in **Figure (3.7.3)** outlines the steps through which the parent entity registers a child. Initially, the parent accesses the "Add Child" process using the AddNewChild (Name, Age) method to submit the child's details. This information is then passed to the "Validate Child Information" process through the validate (Name, Age) method. An alternative flow (alt box) is introduced to handle conditional outcomes: if the child's information is valid, the process continues to store the data in the database. If the information is invalid, an error message is displayed to the parent using the Error (InvalidEntry) method, allowing the parent to correct the entry before proceeding.

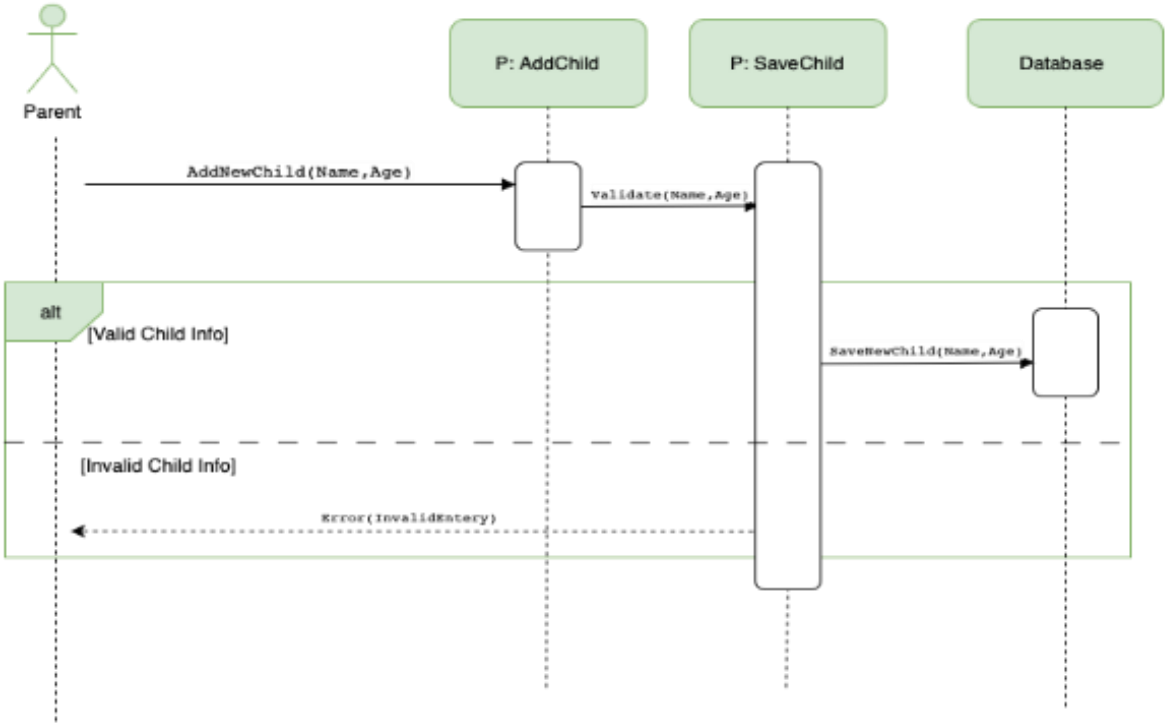


Figure 3.7.3: BVN -Add new child sequence diagram

3.7.4 Assign tasks

The sequence diagram for assigning tasks in **Figure (3.7.4)** describes the process through which a parent can assign either an existing task or create a new one. Initially, the parent encounters an alternative flow (alt box) where they choose between two paths: creating a new task or assigning an existing one.

- If the parent opts to create a new task, they enter the "Create Custom Task" process using the CreateTask(Name) method. The process then moves to the "Set Deadline" step outside the alt box using the DueDate(Date, Time) method, and the task is saved into the database with the Save(Task, Due) method.
- If the parent chooses to assign an existing task, they proceed to the "Assign Task" process, using the customTask(taskName, Description) method to select the task. Similar to the previous path, the process continues to the "Set Deadline" step outside the alt box via the DueDate(Date, Time) method, and the task is saved into the database using the Save(Task, Due) method.

In both cases, the task assignment is completed once the task and deadline are stored in the database.

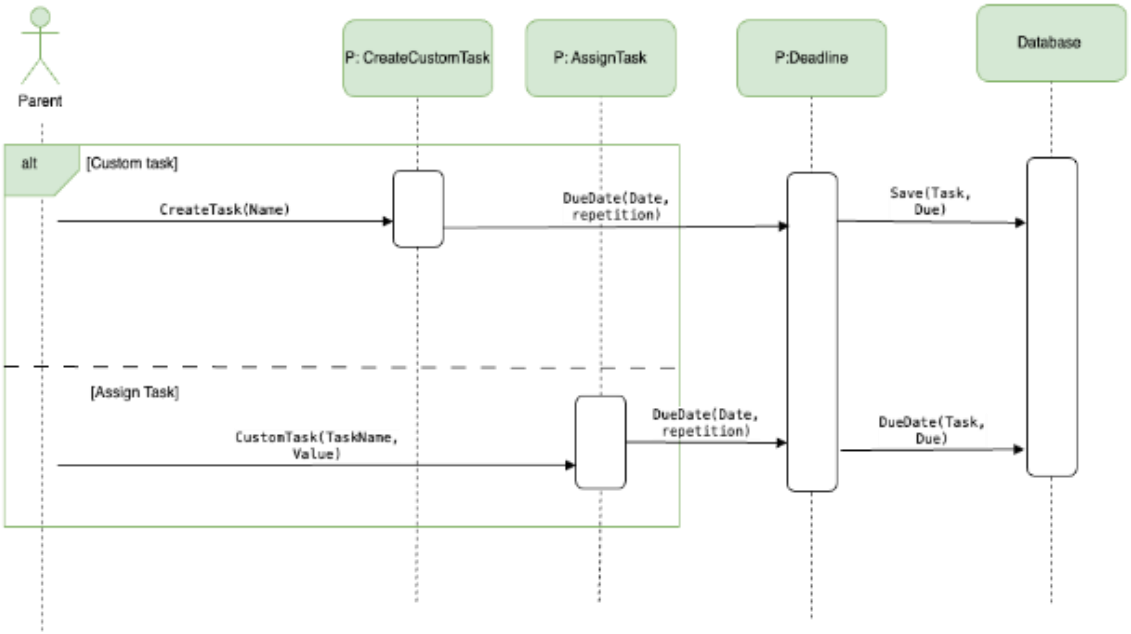


Figure 3.7.4: BVN - Assign task sequence diagram

3.7.5 Rewards

This sequence diagram in **figure (3.7.5)** demonstrates the flow of how rewards are assigned to tasks within the system. The process begins with the parent selecting a task and displaying the page for reward assignment. The parent has the option to either choose a built-in reward or create a custom reward. The system then acknowledges the task and assigns the selected reward to the task. Once the reward is linked to the task, the system saves the reward and sends a confirmation message, indicating that the process has been successfully completed. This diagram highlights the flexibility of reward assignment, either through predefined options or custom inputs.

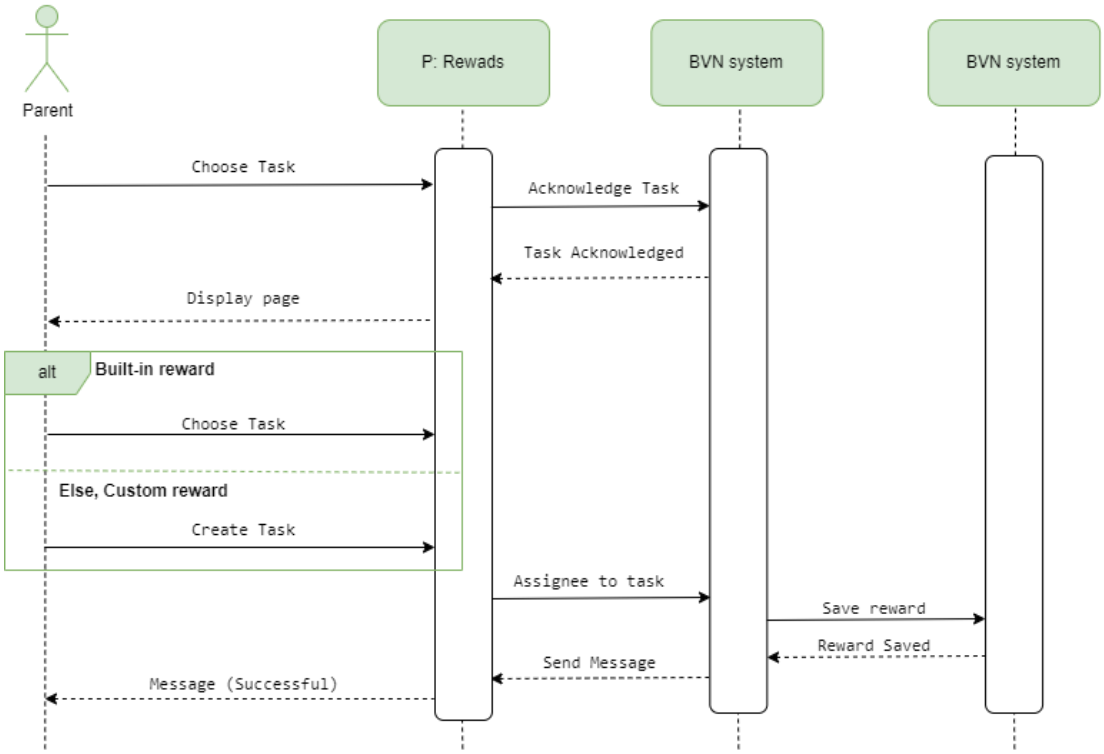


Figure 3.7.5: BVN -Rewards sequence diagram

3.7.6 Check task

The sequence diagram for "Check Task" in **Figure (3.7.6)** illustrates the process by which a Child and Parent interact with the BVN system to manage assigned tasks. It begins with the Child displaying their assigned tasks from the Parent's to-do list. The Child then opens the task list and marks a specific task as completed. This action prompts the BVN system to update the task status. The system confirms the update, ensuring that both the Child and Parent are aware of the changes made. This diagram effectively captures the flow of information and the steps taken to track task completion within the system.

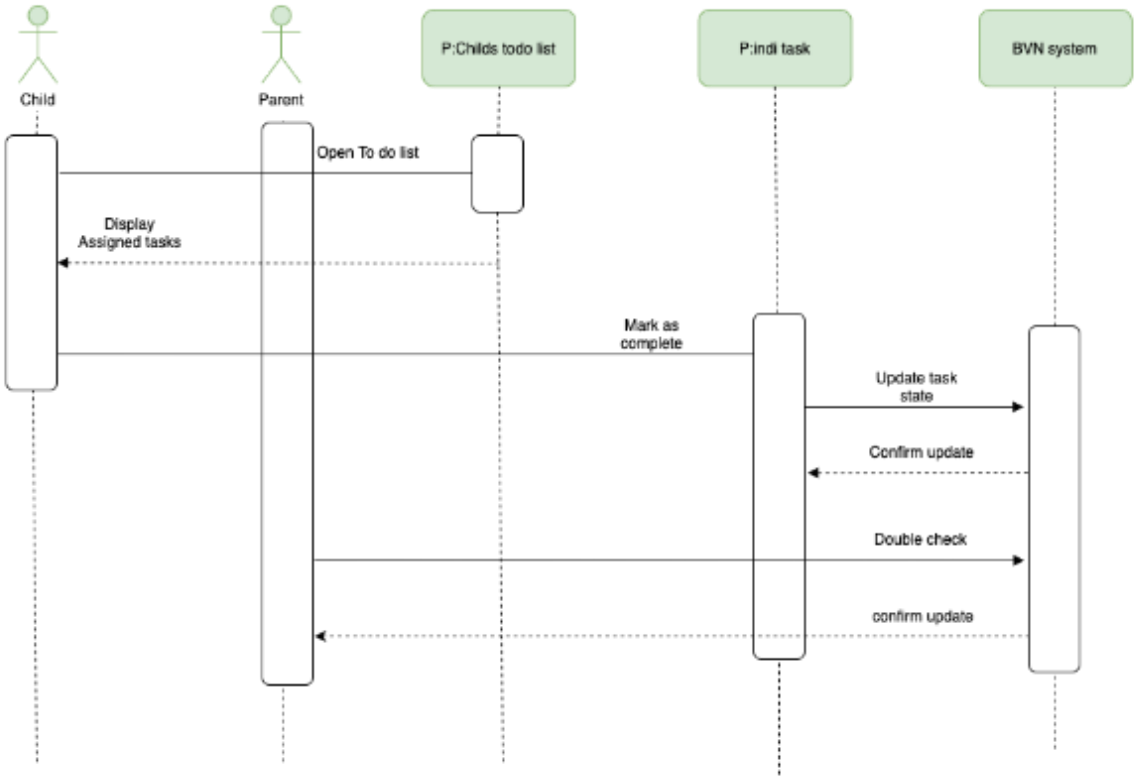


Figure 3.7.6: BVN -Check task sequence diagram

3.7.7 Monitor Progress

The sequence diagram for "Monitor Progress" in **Figure (3.7.7)** outlines the interactions between the Child, Parent, BVN system, and Database during the reward process. The sequence begins with the Child requesting a reward from the Parent. The Parent then selects either a built-in reward or a custom reward. Depending on the choice, the system processes the request and updates the inventory accordingly. Once the task is completed, the BVN system notifies the Parent, ensuring that all parties are informed of the progress. This diagram effectively illustrates the conditional flow and communication between the components involved in monitoring and rewarding tasks.

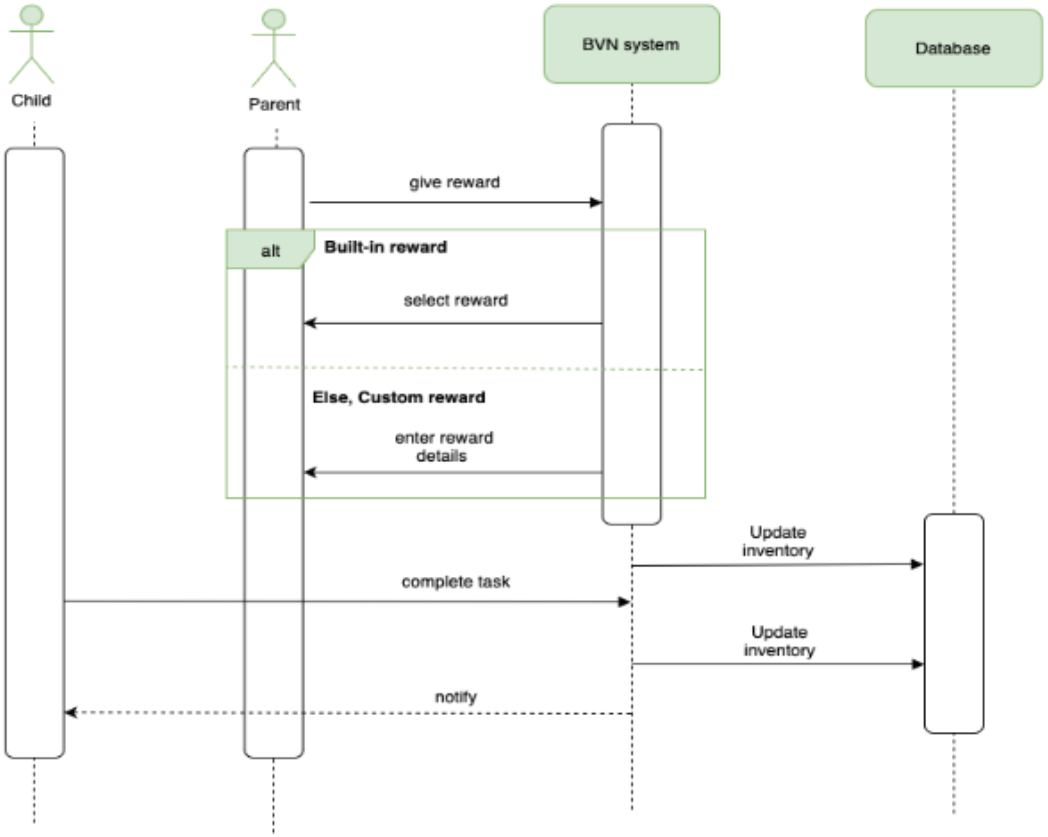


Figure 3.7.7: BVN- Monitor progress sequence diagram

3.7.8 Challenges

This sequence diagram in **figure (3.7.8)** outlines the process for creating challenges in the system. The parent initiates a new challenge, and the system displays the challenge configuration page. The parent enters the necessary details, adds children to the challenge, and clicks save. The system processes the information, resaves the command, and sends the challenge data to the database. Once the challenge is successfully saved, a confirmation message is sent back to both the parent and child. This flow emphasizes the collaborative aspect of the challenge creation, involving both the parent and system for effective task management and goal setting.

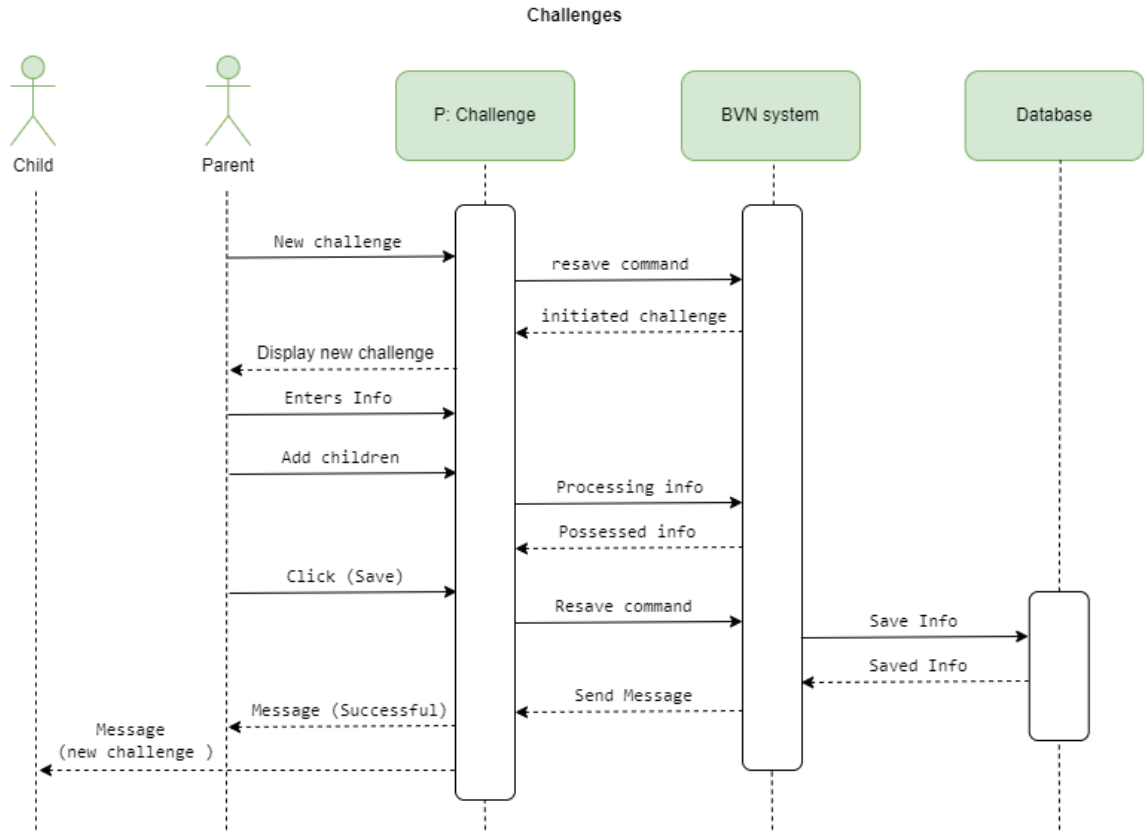


Figure 3.7.8: BVN -Challenges sequence diagram

3.7.9 view inventory

The sequence diagram in **Figure (3.7.9)** illustrates the process of viewing inventory within a system. It begins with a "Child" initiating a request to open the inventory. This request is sent to the BVN system, which then interacts with the Database to fetch the inventory data. The diagram includes a conditional flow: if the inventory is empty, a message indicating that no items are available is displayed; otherwise, the system lists the received inventory items for the user. This visual representation effectively captures the interactions and decision-making involved in the inventory viewing process.

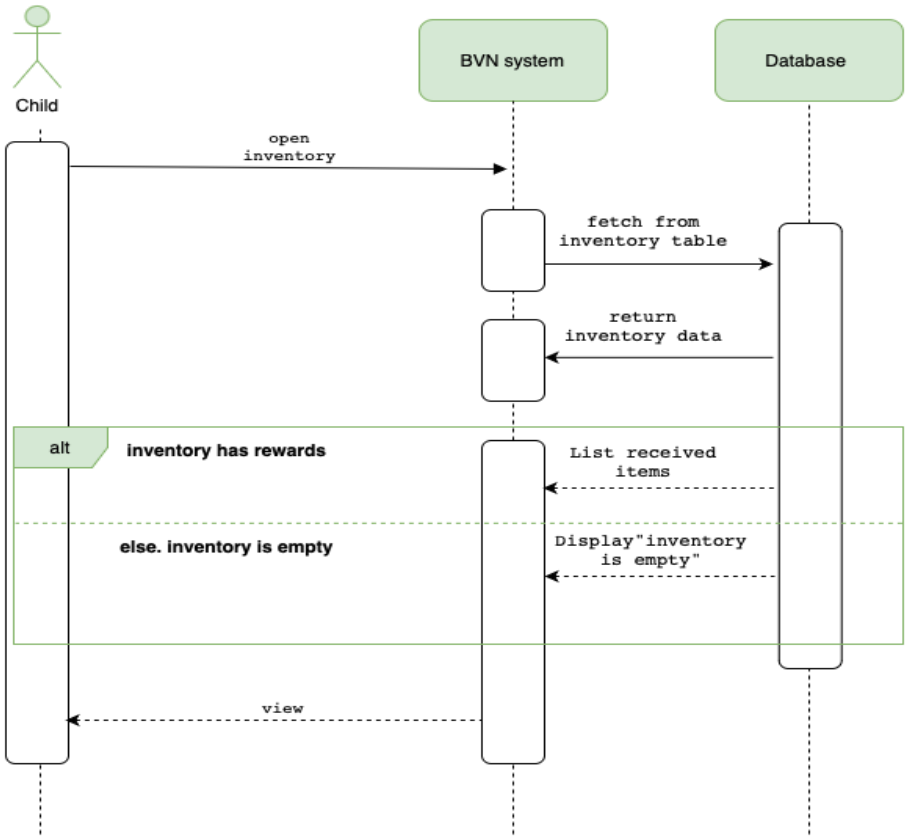


Figure 3.7.9: BVN -View inventory sequence diagram

The architecture and design of the BVN Batal-Value Nest application have been thoroughly examined in this chapter. Through an analysis of the project's environments, class diagrams, and design features, we have developed a comprehensive knowledge of the application's structural operation. A strong and captivating user experience is made possible by this foundation, which guarantees that the app is well-designed to satisfy user needs while preserving efficiency and scalability.

Chapter IV: Prototype

This chapter focuses on the development of the prototype for the BVN Batal - Value Nest application, offering a preview of its key features and user experience. The prototype highlights how parents and children can interact with the app, covering functionalities such as task assignment, progress tracking, and rewards.

The thoughtful color scheme, which was chosen to provide a nurturing and stimulating atmosphere for kids, is a crucial component of the design. The color scheme is intended to promote feelings of serenity, concentration, and optimism, which keeps kids calm and interested while they use the app. Certain hues are linked to development and success, which encourages advancement and sustained involvement. Others encourage children to actively engage with the tasks and rewards by igniting their enthusiasm and energy. By integrating these components, the app creates a warm and engaging environment that encourages the development of habits in a lighthearted and aesthetically pleasing manner.

Unity was also chosen as the development platform because of its adaptability and user-friendliness. Unity provides a powerful platform for creating dynamic and interactive experiences across various devices, ensuring lasting engagement for children using the app. Moreover, rapid development and testing are made possible by its user-friendly interface, allowing simpler iterations and enhancement of any features in response to user feedback. Additionally, Unity's broad support for animations, 2D graphics, and interactive elements made it suitable for creating the entertaining and instructive experience that is intended for the app, ensuring both parents and children will find it simple to use and entertaining.

4.0 BVN Home page:

The homepage of the **BVN App** serves as the entry point for children and parents, providing easy access to value instilled habit-tracking, rewards, and progress insights. It sets the foundation for an engaging experience, reinforcing positive values through a user-friendly design. A well-structured homepage keeps users motivated, making the value-building process an enjoyable journey.

Figure (4.1) shows the main interface of the "BATAL Value nest" platform which presents two primary options, "Create New Account" and "Login," which are prominently displayed, allowing users to easily access the platform's functionalities.



Figure 4.1: Interface – BVN Home page

4.1 Registration page:

Users can register a new account via the registration page. The form asks for first and last names, email address, password, and mobile number. Additionally, a dropdown menu allows users to choose their role, such as "Mother" or "Father." Moreover, a Finish button at the bottom indicates the completion of the sign-up procedure, as illustrated in **Figure 3.7.1**.



Figure 4.2: interface - BVN Registration page

4.2 Log-in page:

Users with existing accounts are the target audience for this page. After entering the email address and password, a **"Log In"** button allows users to access their account. as illustrated in **Figure 3.7.2.**



Figure 4.3BVN Interface – BVN Log-in page

\

4.3 Parent account page:

This page shows the parent's account dashboard, which includes personal information like name, phone number, email, and role. The lower section shows the list of children associated with the parent, along with their custom avatars with an add child option for up to 5 children



Figure 4.5: Interface – BVN Parent account page

4.4 Settings:

The **"Settings"** page of the application is designed with a **child-friendly** theme, featuring cartoon illustrations and soft colors to create an engaging and comfortable experience.

Editing the personal profile, managing linked accounts, modifying privacy and security settings, notifications, language preferences, app details, and logging out are just a few of the options available through the vertical menu on the left.



Figure 4.5: Interface - BVN Settings

4.5 Create a user for a child:

As shown in **Figure (4.6)**, this page allows users to add a child's profile to the application. It features a playful and inviting design with soft pastel colors and cartoon elements to appeal to young users. The interface includes a customizable avatar, a text field for entering the child's name, and a slider to set the child's age. The "إضافة" (Add) button is clearly placed to submit the information. The layout is intuitive and simple, making it easy for parents or guardians to create and manage their child's profile. as illustrated in **Figure 3.7.3.**



Figure 4.6: Interface - BVN Add child.

4.6 Childs Overview page:

Figure (4.7) shows the overview of the child's daily tasks and rewards. It is designed with a vibrant, child-friendly theme, featuring character illustrations and a rewards system to motivate children. Tasks are displayed in card formats, with some marked as completed ("منجزة") and others as pending. The top section shows the child's avatar and name, while the middle section lists the daily tasks, such as brushing teeth or playing, that the child needs to complete to earn rewards. At the bottom, the rewards system is represented visually, including badges and tickets, to encourage engagement through positive reinforcement.

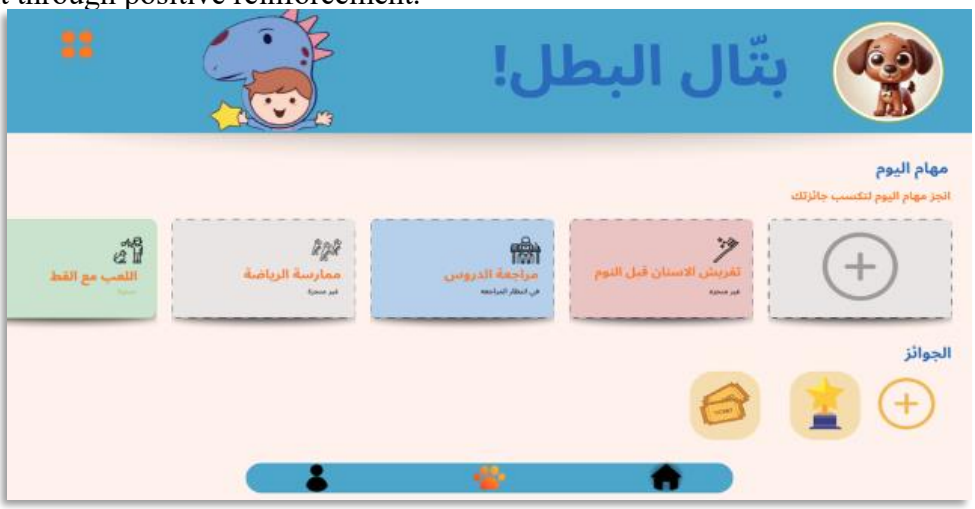


Figure 4.6: Interface- BVN Child's Overview Page

4.7 Assign tasks:

The parent can easily assign tasks to their children by choosing one of the options as shown in the **Figure (4.8)**. it gives freedom to the parent to either chose the built-in tasks or create a custom one. as illustrated in **Figure 3.7.4**

.



Figure 4.8: Interface – BVN Assign tasks

In case a parent decides to create a custom task, the prompt shown in **Figure (4.8.1)** will be displayed to allow the parent to fill in the task's specifications such as the task title, targeted values, duration or reptation which may be repeated daily, weekly or monthly.



Figure 4.8.1: Interface – BVN Create task

A default list of built in tasks will be displayed to the parent as in **Figure (4.8.2)**, and the parent can choose from this list by checking it or view the tasks details by taping on the task itself. (**Figure (4.8.3)**).



Figure 4.8.2: Interface – BVN Choose task



Figure 4.8.3: Interface – BVN Choose task

4.8 Give rewards:

By clicking on the rewarding option, the prompt in **Figure (4.9)** will be displayed to the parent to fill in the reward's title, reason of earning this award and an option to choose a specific task to be mapped to. and finally, an icon to be displayed. The parent can choose either to upload an image to be set as an icon or to select one of the built-in icons. as illustrated in **Figure 3.7.5.**

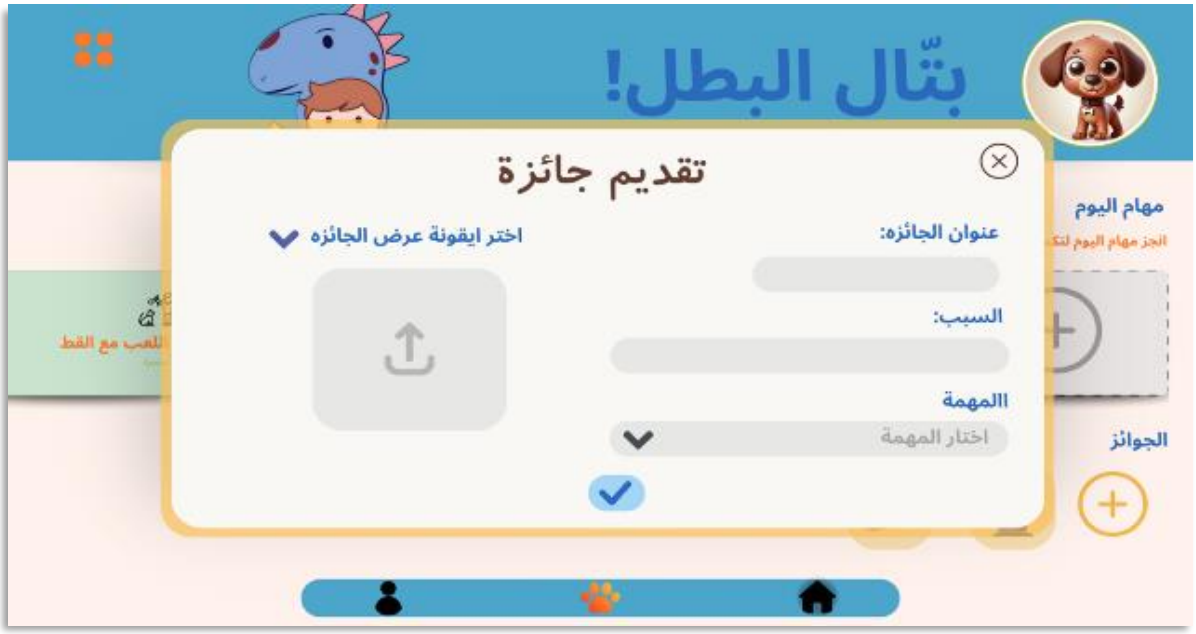


Figure 4.9: Interface – BVN Give rewards

4.9 Initiate challenges:

This interface in **figure (3.10)** allows parents or administrators to create customized challenges tailored to individual children. The design enables the selection of specific tasks and assign values, to ensure children of different ages engage in fair competition by earning equivalent value points based on their efforts. Users can select children from the avatar list on the left, input challenge titles, choose task values, and set challenge durations. The easy-to-use layout, featuring dropdown lists, checkboxes, and date selection, ensures a smooth setup process. The flexibility encourages meaningful engagement, as parents can assign age-appropriate tasks while maintaining fairness through value-based rewards. as illustrated in **Figure 3.7.8.**



Figure 4.10: Interface – BVN Initiate challenge

4.10 Child-Log in page:

The interface portrayed in **Figure (4.10)** represents the **child login page** for the BVN application. It provides a playful and engaging experience for young users, featuring pet avatars which represent individual children linked to the parent's account. This makes it a user-friendly layout that allows children to select their profile easily by tapping on their assigned avatar, making the login process intuitive and enjoyable. The icons for navigation are simple and clear, with arrows enabling easy scrolling through multiple profiles if needed.



Figure 4.11: Interface -BVN Child-Log in page

4.11 Childs home page:

This interface shown in **Figure (3.11)** serves as the central dashboard for children to track their tasks, achievements, and progress. It features four main sections: "Challenges," "Today's Task List," "Achievement Inventory," and "Progress Tracker," each represented with child-oriented illustrations to foster engagement. On the right, the child's profile is prominently displayed, showing the pet avatar (in this case, a cat), along with the child's name, age, and earned points. The neat, visual breakdown makes it easy for children to navigate between tasks and track their progress while maintaining motivation using rewards and goals. This structure aligns with the app's goal of habit tracking and gradual progress-building.

Additionally, there is an option for the child to change the interface theme from the available designs, allowing for a personalized experience that enhances engagement. This structure aligns with the app's goal of habit tracking and gradual progress-building



Figure 4.12: Interface – BVN Child home page

4.12 To do list page:

Figure (4.12) shows the task tracker that aims to encourage responsible behavior and fosters honesty in a child. The tracker lists various daily tasks assign by the parent previously, which the child is expected to complete.. The design allows the child to check off each task as it is accomplished, causing the corresponding box to turn green. This initial checkbox serves as the child's self-report of task completion. The parent then has the opportunity to review and confirm the child's progress by marking the adjacent blue box. If the parent verifies that the task was indeed completed, the second box will turn green, and the child will be rewarded, through a system of virtual currency (e.g., coins). Conversely, if the parent determines the task was not finished, the box will turn red, and the child will face a consequence, such as losing previously earned rewards as the value of being truthful or trustworthy has been broken. Overall, the task tracker is a tool designed for promoting responsible behavior and positive values in the child's daily routine.



Figure 4.13: Interface – BVN To-Do list page

4.13 Prayer tracker page:

The interface illustrated in **Figure (4.13)** presents the five daily prayers, enabling both children and parents to effectively monitor their completion, and motivate them to stay consistent



Figure 4.13: Interface- BVN Prayer tracker page

4.14 Challenges page:

This interface portrayed in **Figure (3.13)** presents a racing game where children engage in challenges by earning values. The game is designed to allow children of different ages to participate fairly by completing distinct tasks with equivalent value points. The screen displays animated cars racing along lanes, each linked to a child profile represented by their avatar on the right. A crown on the top avatar indicates the current leader based on accumulated points. The arrows on both sides provide easy navigation, while the playful graphics encourage engagement. This gamified approach motivates children to complete tasks, promoting healthy competition regardless of their age or task differences.



Figure 4.14: Interface – BVN Challenge page

4.15 Progress page:

Figure (4.14) illustrates the child's progress across various character values. Each value, represented by a specific category as a trophy, advances as the child completes associated tasks. Progress is not only driven by task completion but also through parental involvement. For instance, when parents review and confirm task completion, it contributes to increasing the progress bars for attributes such as honesty and responsibility. This design encourages consistent development by integrating both individual efforts and parental validation. as illustrated in **Figure 3.7.7.**



Figure 4.15: Interface – BVN Progress page

4.16 Rewards inventory:

Rewards inventory page allows the child to browse all achievements they have earned as shown in **Figure (4.15)**. Each accomplishment is displayed along with the date it was obtained and the reason behind it. For example, the "Responsibility Hero" award was achieved by successfully completing tasks with full responsibility, while a -custom reward- cinema trip reward was granted for helping a sibling. This feature reinforces positive behavior by providing a visual record of progress and rewards, motivating the child to continue building good habits and values. as illustrated in **Figure 3.7.9.**



Figure 4.16: Interface – BVN Reward inventory

To bring the BVN Batal-Value Nest application to life, the prototype described in this chapter is essential because it provides a hands-on example of its main functions and overall user experience. The important interactions with this prototype have been visualized, including task assignment, progress tracking, rewards, and children's engagement through interactive avatars. The design decisions, especially the thoughtfully chosen color palette, are crucial in promoting the child's development of new skills and habits. In addition to the visual appeal, the colors were picked for their psychological effects, which include fostering motivation, focus, and serenity while preserving a feeling of vitality and enjoyment. This careful blend in design components guarantees that the application stays interesting and user-friendly, enhancing both its usability and its effectiveness in fostering positive habits and values.

Chapter V: BVN Implementation

This chapter describes the implementation of the Project, outlining the technical details, tools, and methodologies used to realize the application. The focus was on translating the design into a functional product that meets the requirements of Arabic-speaking communities while promoting values through a gamified experience. The implementation aligns with the project goals by ensuring that every component contributes to fostering positive behaviors and values in children, making the application not only a tool for habit formation but also a platform for cultural engagement.

5.1 Development Environment and Tools

The development of the BVN application utilized a range of tools and technologies to ensure optimal performance and user engagement. The primary platform chosen for development was the Unity Engine, utilizing C# for scripting. The Visual Studio IDE was employed for coding, providing a robust environment for development.

For database management, MySQL was selected, paired with PHP for server-side scripting to handle data interactions. This combination was chosen for its reliability and scalability. enabling efficient data storage and retrieval.

5.2 System Architecture and Frameworks

The system architecture of the BVN Project was designed with a modular approach to enhance maintainability and scalability. The Unity Engine served as the frontend, providing a visually engaging interface while managing user interactions and animations. The backend, built with MySQL and PHP, facilitated data management, and ensured real-time updates.

The architecture followed a client-server model, allowing seamless communication between the Unity application and the database. This setup ensures that user data, tasks, and rewards are efficiently managed, providing a smooth user experience.

5.3 Database Implementation

For data storage, MySQL was chosen due to its robustness and ability to handle complex queries efficiently. The database was structured to store user profiles, tasks, rewards, and progress, ensuring quick access and retrieval of information.

Security measures included user authentication and data encryption to protect sensitive information. The database interacted with the Unity application through PHP scripts, which managed requests and responses, ensuring a secure and efficient data flow.

5.4 Core Functionalities

The core functionalities of the BVN application were implemented to promote user engagement and facilitate habit formation. Parents could assign tasks to their children, which were tracked through a gamified progress system that provided visual rewards, such as stars and badges, for completed tasks. Additionally, the application was designed to support the idea of instilling positive values in children through their completed tasks. By engaging in various activities, children not only developed healthy habits but also learned important values such as responsibility, perseverance, and teamwork.

Challenges faced during implementation included ensuring the smooth integration of these gamification elements and developing a user-friendly interface that appealed to children. One of the key challenges was designing a system that effectively reinforced positive behaviors while keeping the experience fun and engaging. It was essential to strike a balance between entertainment and educational value, ensuring that the reward system motivated children without overshadowing the importance of the values being cultivated.

5.5 Gamification Techniques

Gamification was a central aspect of the Value Nest Project, employing various game elements to engage children and encourage them to complete value-based tasks. Rewards and levels were implemented to motivate children, with algorithms designed to evaluate progress based on task completion and engagement levels.

The visual design of gamified elements included animations and leaderboards, creating a competitive yet friendly environment that fostered intrinsic motivation among users.

5.6 User Interface (UI) and User Experience (UX)

The UI/UX design focused on creating a child-friendly interface, ensuring that children aged 6-9 could navigate the app easily. Bright colors, large icons, and intuitive layouts were incorporated to usability and engagement.

Accessibility features were also included, such as simple navigation and Arabic language support. This thoughtful design approach aimed to make the application not only functional but also enjoyable for children and parents.

Chapter VI: System validation

System validation is essential to ensure that the BVN application effectively supports habit formation and value development in children while maintaining a smooth user experience. This phase involved testing the app's core functionalities, including task assignment, gamification elements, and parent-child interaction. Usability tests were conducted to assess engagement, accessibility, and ease of navigation for both parents and children. Additionally, technical evaluations ensured performance stability and data security. The results of this validation process provided valuable insights, guiding necessary refinements to enhance the app's effectiveness and overall user satisfaction.

6.1 Test Approach/Methodology

The BVN application was validated using a structured testing approach combining functional and non-functional testing. The testing strategy incorporated black box testing for functionality validation and User Acceptance Testing (UAT) for subjective evaluation. Black-box testing focused on verifying system outputs against specified inputs without examining internal code structures, ensuring the application meets its functional requirements. UAT was conducted by involving parents and children to ensure the app's usability, cultural relevance, and overall user experience.

The testing process included a group of 10 participants, consisting of both parents and children, who actively engaged in evaluating the application. The feedback from the participants was overwhelmingly positive, with users expressing high satisfaction regarding the app's ease of use, engaging features, and effectiveness in promoting positive habits.

Testing cycles were iterative, allowing issues to be identified, corrected, and retested to ensure reliability and performance stability. All tests were documented, and results were compared against success criteria outlined in the requirements. Additionally, based on user feedback, minor enhancements were implemented to further optimize the experience, reinforcing the app's usability, accessibility, and overall functionality.

6.2 Test Plan

A well-structured test plan is essential for ensuring the reliability and functionality of the application. This section defines the **scope of features to be tested and the features not to be tested , the testing tools and the data** used in the testing process, outlining the identification and addressing potential issues.

6.2.1 Features to be Tested

The testing process focuses on evaluating key features to ensure the app functions smoothly and meets user needs. This includes user registration and login, Task management , etc .

- **User Registration and Login**: Validation of parent and child account creation, including data validation and error handling.
- **Task Management**: Assigning, viewing, and completing tasks, with parent and child-specific interfaces.
- **Reward System**: Assigning custom and built-in rewards, tracking earned rewards in the child's inventory.
- **Progress Monitoring**: Display of progress charts and task completion summaries for parents and children.
- **Challenge Feature**: Creation, participation, and result tracking of competitive or cooperative challenges.
- **Interactive Pet Avatar**: Engagement with the avatar for reminders and motivational support.
- **Localization**: Arabic language support across all features and interfaces.

6.2.2 Features not to be Tested

Some components were not tested, as the app is still in its infancy, namely:

- **Third-Party Authentication APIs**: Assumed reliable if external service integration is verified.

- **Payment Gateway Integration:** Placeholder for future versions, excluded from current scope.

6.2.3 Testing Tools and Environment

- **Testing Tools**:
  - Manual testing through test scripts.
  - Automated test cases using Unity Test Framework for unit tests and Play Mode tests.
- **Environment**:
  - Devices: Android and iOS smartphones.
  - Platforms: Unity for development, MySQL for database management.
  - Localization settings enabled for Arabic.

6.3 Test Data

The test data included:

- Sample parent and child accounts with varying names, emails, and passwords.
- Predefined tasks (e.g., "Brushing Teeth") and custom tasks with varying frequencies.
- Rewards data, both built-in and custom.
- Progress metrics for multiple task completion scenarios.
- Arabic text samples to validate localization support.

6.4 Test Cases

Test Case ID	Test Description	Expected Outcome	Result
TC-01	Register a parent account with valid details	Account created successfully	**Passed**
TC-02	Register a parent account with missing email	Error message displayed	**Passed**
TC-03	Log in with correct credentials	Access granted	**Passed**
TC-04	Log in with incorrect password	Error message displayed	**Passed**
TC-05	Add a child profile	Child added to parent account	**Passed**
TC-06	Assign a task to a child	Task appears in child's to-do list	**Passed**
TC-07	Mark a task as completed	Task status updated	**Passed**
TC-08	Assign and track a reward	Reward added to inventory	**Passed**
TC-09	Monitor child progress	Progress chart displays correct data	**Passed**
TC-10	Create and manage a challenge	Challenge status updated correctly	**Passed**

6.5 Test Result

The overall test results indicated that the BVN application met its core functional and non-functional requirements. Key highlights include:

- **Success Rate:** 100% of high-priority test cases passed.
- **User Feedback:** Positive feedback from user acceptance testing indicated high engagement, usability, and cultural relevance.
- **Performance Metrics:** The application met performance benchmarks for loading time and responsiveness.

Issues identified during initial testing, such as minor localization inconsistencies and reward assignment edge cases, were resolved through iterative refinements. The system is validated as ready for deployment, supporting the objective of fostering positive habits and values in children through engaging, culturally tailored interactions.

Chapter VII: Conclusion and future work

In a world where digital experiences dominate children's engagement, the BVN (Batal Value Nest) project emerged as a culturally sensitive and value-oriented solution to the challenges parents face in instilling core values and positive habits. By integrating cultural alignment, gamification, and value-based tasks, BVN provides a unique platform that fosters both habit formation and moral development in children aged 6 to 9.

The project's approach, built upon parental insights, ensures that the app resonates with its target audience by combining educational engagement with culturally relevant content. Core features such as task assignment, progress tracking, and reward mechanisms have been tailored to Arabic-speaking families, addressing the dual needs of habit-building and value enrooting.

BVN has laid the foundation for a meaningful tool that supports families in building a new generation of responsible, empathetic, and value-driven individuals. However, the journey does not end here, as the evolving needs of children, families, and technology demand continuous innovation and improvement.

Future Work

As we move forward, there are several suggested directions to enhance the BVN platform and expand its impact such as:

1. **Incorporating** different **Games to build values into BVN not just tasks**
2. To ensure children remain engaged and motivated, future versions of the app will include interactive games within the Batal Value Nest environment.

3. **Introducing More Themes**
   The app's user experience will be enriched by introducing a wider variety of themes to cater to diverse interests among children. This customization will enhance user engagement and allow children to feel a greater sense of ownership over their learning and growth journey.

4. **Expanding Challenges to Include Friends**

   Currently, the app focuses on challenges and competition among siblings. In the future, we aim to broaden this scope to include friends, enabling children to collaborate and compete within their wider social circles. This feature will encourage teamwork and build a sense of community among young users.

5. **Support for Multiple Operating Systems**

   The current focus has been on Android devices. To ensure wider accessibility, future iterations of BVN will support additional platforms such as Android, allowing more families to benefit from the application regardless of their preferred operating systems.

Final conclusions:

The BVN project is a significant step toward integrating technology, cultural values, and habit formation into a single platform that meets the needs of modern families. By expanding the platform's features, themes, accessibility, and interactivity, BVN will continue to evolve as a comprehensive tool for nurturing responsible, empathetic, and value-driven children in an ever-changing world. Through innovation and dedication, we aim to make BVN a cornerstone of value-based education in the digital age.

SOURCE CODE:

To find the source code, please refer to the references below, where it is available as a Google Drive link.

References

App References

- Habitica: Gamified Taskmanager. (n.d.). *Apple App Store*. Retrieved from https://apps.apple.com/sa/app/habitica-gamified-taskmanager/id994882113
- Supers: Tasks & Rewards. (n.d.). *Apple App Store*. Retrieved from https://apps.apple.com/sa/app/supers-tasks-rewards/id6444745824

Website Reference

- First Habits. (2024, April 9). *Navigating the digital landscape: Cultivating positive technology habits in children.* Retrieved from https://firsthabits.com/technology/

Journal Articles

- Chen, Z. (2023). The influence of school's reward systems on students' development. *Journal of Education, Humanities and Social Sciences, 8*, 1822-1827. University of California, Irvine.
- Hirsch, L., & Milligan, L. (2020). Developing children's social and emotional skills through digital games. *Computers & Education*.
- Kalogiannakis, M., Papadakis, S., & Zourmpakis, A.-I. (2021). Gamification in science education: A systematic review of literature. *Education Sciences*.
- Lee, H., & Wong, K. (2022). Gamification in child-centered applications. *Educational Games Review*.
- Park, J., & Kim, H. (2021). Gamification in mobile applications for children's education: A systematic review. *Journal of Educational Research and Innovation*.
- Wang, Y., & Chen, X. (2022). Motivating habit formation in children using gamified apps: A case study of value-driven mobile education. *Interactive Learning Environments*.

Source code

- https://drive.google.com/drive/folders/1x3a5Qy_cTsfEcLxGwdk2dVZRDvktt9pD?usp=drive_link

Appendix -1

Survey:

a. **Pre survey (before evaluation):**

هدف هذا الاستبيان إلى فهم آراء الآباء وتفضيلاتهم بشأن الحلول التكنولوجية المستقبلية التي يمكن أن تدعم تطوير قيم أطفالهم وعاداتهم. يركز مشروع "بطل" على إنشاء تطبيق محمول باللغة العربية مصمم لمساعدة الأطفال الذين تتراوح أعمارهم بين ٦-٩ سنوات على تنمية العادات الإيجابية والقيم الأساسية من خلال تجارب تفاعلية وممتعة. من خلال دمج المهام المرتكزة على القيم والميزات التفاعلية، يسعى التطبيق إلى تمكين الآباء في توجيه أطفالهم نحو نمو سلوكي ذي مغزى.

This survey seeks to understand parents' views and preferences on future technological solutions that can support the development of their children's values and habits. The project, "BVN Batal - Value Nest," focuses on creating an Arabic-language mobile application designed to help children aged 6-9 cultivate positive habits and core values through engaging, gamified experiences. By integrating value-driven tasks and interactive features, the app aims to empower parents in guiding their children toward meaningful behavioral growth.

1. What age group does your child belong to?
ما هو عمر طفلك؟
   - 6-7 years - 6-7 سنوات
   - 8-9 years - 8-9 سنوات

2. How often do you use mobile applications for your child's learning?
ما مدى استخدامك للتطبيقات المحمولة في تعليم طفلك؟
   - Daily - يومياً
   - Weekly - أسبوعياً
   - Monthly - شهرياً
   - Rarely - نادراً
   - Never - أبداً

3. What values do you consider most important to instill in your child? (Select all that apply)?
ما القيم التي تعتبرها الأكثر أهمية لتعليمها لطفلك؟ (اختر كل ما ينطبق)؟
   - Empathy - التعاطف
   - Responsibility - المسؤولية
   - Cooperation - التعاون
   - Honesty - الصدق
   - Respect - الاحترام

4. Have you used any habit-tracking apps before?
هل استخدمت أي تطبيقات تتبع العادات من قبل؟
   - Yes - نعم
   - No - لا

5. What features would you find most useful in a habit-tracking app?(Select all that apply)
ما الميزات التي تجدها أكثر فائدة في تطبيق تتبع العادات؟ (اختر كل ما ينطبق)؟

- Task assignment - تعيين المهام
- Rewards system - نظام المكافآت
- Progress tracking - تتبع التقدم
- Interactive elements (e.g., avatars) - عناصر تفاعلية (مثل الأفاتار)
- Parental controls - تحكمات الوالدين

6. How important is it for you that the app is available in Arabic?
ما مدى أهمية أن يكون التطبيق متاحًا باللغة العربية بالنسبة لك؟

- Very important - مهم جداً
- Important - مهم
- Neutral - محايد
- Not important - غير مهم
- Not at all important - ليس مهمًا على الإطلاق

7. What challenges do you face in instilling values in your child? (Open-ended)
ما التحديات التي تواجهها في غرس القيم في طفلك؟ (مفتوح)

b. Pre survey (before evaluation):

يهدف هذا الاستبيان إلى فهم آراء الآباء وتفضيلاتهم بشأن الحلول التكنولوجية المستقبلية التي يمكن أن تدعم تطوير قيم أطفالهم وعاداتهم. يركز مشروع "بطل" على إنشاء تطبيق محمول باللغة العربية مصمم لمساعدة الأطفال الذين تتراوح أعمارهم بين ٦-٩ سنوات على تنمية العادات الإيجابية والقيم الأساسية من خلال تجارب تفاعلية وممتعة. من خلال دمج المهام المرتكزة على القيم والميزات التفاعلية، يسعى التطبيق إلى تمكين الآباء في توجيه أطفالهم نحو نمو سلوكي ذي مغزى.

This survey seeks to understand parents' views and preferences on future technological solutions that can support the development of their children's values and habits. The project, "BVN Batal - Value Nest," focuses on creating an Arabic-language mobile application designed to help children aged 6-9 cultivate positive habits and core values through engaging, gamified experiences. By integrating value-driven tasks and interactive features, the app aims to empower parents in guiding their children toward meaningful behavioral growth.
behavioral growth.

1. What age group does your child belong to?
ما هو عمر طفلك؟

- 6-7 years - 6-7 سنوات
- 8-9 years - 8-9 سنوات

2. How often do you use mobile applications for your child's learning?
ما مدى استخدامك للتطبيقات المحمولة في تعليم طفلك؟

- Daily - يومياً

- Weekly - أسبوعياً
- Monthly - شهرياً
- Rarely - نادراً
- Never - أبداً

3. What values do you consider most important to instill in your child?(Select all that apply)?

ما القيم التي تعتبرها الأكثر أهمية لتعليمها لطفلك؟ (اختر كل ما ينطبق)؟

- Empathy - التعاطف
- Responsibility - المسؤولية
- Cooperation - التعاون
- Honesty - الصدق
- Respect - الاحترام

4. Have you used any habit-tracking apps before?

هل استخدمت أي تطبيقات تتبع العادات من قبل؟

- Yes - نعم
- No - لا

5. What features would you find most useful in a habit-tracking app?(Select all that apply)

ما الميزات التي تجدها أكثر فائدة في تطبيق تتبع العادات؟ (اختر كل ما ينطبق)؟

- Task assignment - تعيين المهام
- Rewards system - نظام المكافآت
- Progress tracking - تتبع التقدم
- Interactive elements (e.g., avatars) - عناصر تفاعلية (مثل الأفاتار)
- Parental controls - تحكمات الوالدين

6. How important is it for you that the app is available in Arabic?

ما مدى أهمية أن يكون التطبيق متاحًا باللغة العربية بالنسبة لك؟

- Very important - مهم جداً
- Important - مهم
- Neutral - محايد
- Not important - غير مهم
- Not at all important - ليس مهمًا على الإطلاق

7. What challenges do you face in instilling values in your child? (Open-ended)

ما التحديات التي تواجهها في غرس القيم في طفلك؟ (مفتوح)

8. What are the strategies / approaches you use to install value in your choles (Open-ended)

ما هي الطريقة التي تعزز بها القيم لدى طفلك؟ (مفتوح)

Appendix -1.1

Survey:

c. Post-Survey (After Evaluation)

هدف هذا الاستبانة هو قياس مدى فاعلية تطبيق "بطل" وتأثيره على تطوير عادات الأطفال وقيمهم، بالإضافة إلى تقييم تجربة المستخدمين من حيث السهولة، المتعة، والفائدة العامة. من خلال ملاحظاتكم، يمكننا تحسين التطبيق بشكل أكبر لضمان تجربة تفاعلية فعالة تلبي احتياجات الأطفال وأولياء الأمور على حد سواء.

This survey aims to measure the effectiveness of the "BVN Batal - Value Nest" app and its impact on children's habit development and value formation, as well as evaluate the user experience in terms of ease, engagement, and overall usefulness. Your feedback will help us refine the app further to ensure an effective interactive experience that meets both children's and parents' needs.

1- ما مدى رضاك عن تجربة استخدام تطبيق "بطل" بشكل عام؟ | How satisfied are you with the overall experience of using the "BVN Batal" app?

- راضٍ جدًا | Very satisfied
- راضٍ | Satisfied
- محايد | Neutral
- غير راضٍ | Dissatisfied
- غير راضٍ أبدًا | Very dissatisfied

2- إلى أي مدى ساعد التطبيق في تحسين عادات طفلك الإيجابية؟ | To what extent has the app helped improve your child's positive habits?

- جدًا | Very much
- إلى حد ما | Somewhat
- محايد | Neutral
- لا يساعد كثيرًا | Not much
- لا يساعد أبدًا | Not at all

3- ما مدى جاذبية العناصر التفاعلية والشخصيات الافتراضية لطفلك؟ | How engaging are the interactive elements and virtual characters for your child?

- ممتعة جدًا | Very engaging
- ممتعة | Engaging
- محايدة | Neutral
- غير ممتعة | Not engaging
- غير ممتعة أبدًا | Not engaging at all

4- ما مدى سهولة استخدام التطبيق بالنسبة لك ولطفلك؟ | How easy is the app to use for both you and your child?

- سهل جدًا | Very easy
- سهل | Easy
- محايد | Neutral
- صعب بعض الشيء | Somewhat difficult
- صعب جدًا | Very difficult

5- هل لديك أي اقتراحات لتحسين التطبيق؟ | Do you have any suggestions for improving the app? (مفتوح) |(Open-ended)